Cyclotomic specialization of unicritical multiplier curves and alternating monodromy in period 4

Dongsheng Wei

Abstract

For the unicritical family $f_c(z) = z^d + c$, we introduce a reduced multiplier on its multiplier curves. For cycle period m, with $\gamma = \gcd(m, d-1)$, the usual multiplier is the γ-th power of the reduced one. We prove that every power pullback of the reduced map is geometrically irreducible and that no larger power can be removed from the usual multiplier. Torsion separation and cyclotomic Hilbert irreducibility then give, outside finitely many orders, complete factorization formulas over $\mathbf{Q}$, $\mathbf{Q}^{\mathrm{ab}}$, and fixed number fields. Thus each quadratic period row of the Morton–Vivaldi conjecture has only finitely many exceptions.

We determine all exceptions in period 4: every period-4 delta factor is irreducible over $\mathbf{Q}$. More strongly, for every root of unity $\zeta \neq 1$, the associated sextic over $\mathbf{Q}(\zeta)$ has Galois group containing A_6, remains irreducible over the maximal solvable extension of $\mathbf{Q}(\zeta)$, and has splitting group A_6 there. This yields exact factorizations after solvable base change and the maximal solvable subfields of the associated parameter fields. The generic splitting field is a regular S_6-extension, and all but finitely many torsion specializations retain group S_6 over $\mathbf{Q}^{\mathrm{ab}}$.

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Contents

1 Introduction

Arithmetic questions about the fibers of a finite cover $X \to \mathbf{G}_m$ above roots of unity are governed by two complementary mechanisms. A factorization of the covering function through a power map produces systematic splitting. Once the maximal power has been removed, the pullback condition *(PB)*—geometric irreducibility after every power pullback—allows cyclotomic Hilbert irreducibility to leave only finitely many exceptional torsion fibers. An exact factorization problem therefore has two stages: determine the maximal power obstruction, and then analyze the exceptional fibers that remain.

Multiplier curves in arithmetic dynamics provide a natural setting for both stages. In a polynomial family, a root-of-unity multiplier singles out a parabolic parameter, so delta factors record the arithmetic of torsion fibers on multiplier curves rather than a collection of unrelated specializations.

For the quadratic family $f_c(z) = z^2 + c$, Morton and Vivaldi introduced delta factors whose roots are parabolic parameters of prescribed formal and exact periods, and conjectured that all of them are irreducible over $\mathbf{Q}$ [MV95]. Morton subsequently restated the problem [Mor96, Introduction]. For the correction of a finite-characteristic argument in that article, see [Mor11]. The problem also appears in [Sil07, Exercise 4.12(e)**]. Periods 1 and 2 reduce to cyclotomic irreducibility [MV95, Corollary 3.7], and Koizumi, Murakami, Sano, and Takehira proved the complete period-3 case [KMST25, Theorem 1.2]. Period 4 was the first unresolved complete row.

This paper carries out both stages in three parts. The first places the problem in a general geometric framework for every unicritical family

$$f_c(z) = z^d + c, \qquad d \geq 2.$$

The second proves irreducibility of every period-4 delta factor in the quadratic family. The third strengthens the result to alternating monodromy and stability under arbitrary solvable base change. Thus period 4 is not treated as an isolated low-period calculation: it is the first previously unresolved complete row for which the finite exceptional set left by the general theorem is resolved exactly. The abstract sextic disjointness criterion used in the third part is independent of arithmetic dynamics.

Let

$$\nu_d(m) = \sum_{r|m} \mu(m/r) d^r, \qquad t = d^d c^{d-1}, \qquad \gamma = \gcd(m, d-1).$$

Here μ is the Möbius function, φ is Euler's totient function, Φ_k is the k-th cyclotomic polynomial, and $\mu_\infty \subset \overline{\mathbf{Q}}^\times$ is the group of all roots of unity. For $n \geq 1$, put $\operatorname{rad}(n) = \prod_{p|n} p$, with the empty product equal to 1. Write $D_{d,m}(t,x)$ for the multiplier polynomial in the scaled parameter, and write $\mathcal{P}_{d,m,k}(t)$ for the positive-leading normalization of $\operatorname{Res}_x\big(\Phi_k(x), D_{d,m}(t,x)\big)$. Our first main theorem is summarized as follows.

Theorem 1.1 (Reduced multiplier and eventual factorization)**.** *On the normalization of $D_{d,m}(t,x) = 0$ there is a canonical rational function y such that*

$$x = y^\gamma, \qquad \deg y = \frac{\nu_d(m)}{d\gamma}.$$

The cover y satisfies (PB), and γ is the maximal geometric isogeny degree of x. For every sufficiently large k, the complete factorization of $\mathcal{P}_{d,m,k}$ over $\mathbf{Q}$ is indexed by

$$a \mid \gamma, \qquad (k, \gamma/a) = 1,$$

and the factor indexed by a has degree

$$\frac{\nu_d(m)}{d\gamma}\,\varphi(ka).$$

Over $\mathbf{Q}^{\mathrm{ab}}$ *there are exactly* $\gamma\varphi(k)$ *distinct factors, each of degree* $\nu_d(m)/(d\gamma)$*.*

The new geometric input is the identification of the maximal isogeny. If P denotes the product of the points in a marked m-cycle, then

$$x = d^m P^{d-1}, \qquad y = d^{m/\gamma} P^{(d-1)/\gamma}.$$

At a critical-center point and a point above infinity, y has coprime orders. This proves (PB) by the binomial criterion. Zannier's cyclotomic Hilbert irreducibility theorem then gives full-degree irreducibility over the cyclotomic closure outside finitely many torsion fibers [Zan10, Theorem 2.1]. A torsion-separation argument specific to unicritical dynamics recovers the reduced multiplier from the parameter; this turns the geometric theorem into the exact arithmetic factorization above. The elementary factor count implies

$$\mathcal{P}_{d,m,k} \text{ is eventually irreducible over } \mathbf{Q} \quad\Longleftrightarrow\quad \mathrm{rad}((m, d-1)) \mid k.$$

For $d = 2$ the greatest common divisor is always one, so every fixed row of the Morton–Vivaldi conjecture is irreducible for every sufficiently large k.

The second part gives a uniform, exception-free theorem in period 4. Put $C = 4c$. The period-4 multiplier curve has normalization

$$g(T) = -T^4 - 2T^3 - 4T^2 - 6T + 5 + \frac{8}{T} + \frac{16}{T^2}, \qquad h(T) = -T^2 - 3 - \frac{4}{T},$$

and is parametrized by

$$T \longmapsto \big(g(T), h(T)\big) \tag{1}$$

on $\mathbf{G}_m$. This parametrization is recorded in [KMST25, Remark 2.10] and attributed there to Giarrusso and Fisher [GF95]. In Section 13 we give a direct algebraic proof that the displayed map is the normalization. Pulling cyclotomic polynomials back along the degree-six function g leads, for $k \geq 2$, to

$$\Gamma_k(T) = (-1)^{\varphi(k)} T^{2\varphi(k)} \Phi_k(g(T)).$$

This Laurent expression is in fact a monic polynomial in $\mathbf{Z}[T]$ of degree $6\varphi(k)$.

The exact period-4 result is the following.

Theorem 1.2. *For every integer* $k \geq 2$*, the polynomial* $\Gamma_k(T)$ *is irreducible over* $\mathbf{Q}$*.*

We also strengthen the corresponding fixed-degree statement. For a number field K, write K^{ab} and K^{sol} for its maximal abelian and maximal solvable extensions in a fixed algebraic closure. For a root of unity ζ, set

$$F_\zeta(T) = T^6 + 2T^5 + 4T^4 + 6T^3 + (\zeta - 5)T^2 - 8T - 16.$$

Theorem 1.3 (Exception-free alternating monodromy)**.** *Let* $\zeta \neq 1$ *be any root of unity, put* $K = \mathbf{Q}(\zeta)$*, and let* L_ζ *be the splitting field of* F_ζ *over* K*. Then*

$$A_6 \leq \mathrm{Gal}(L_\zeta/K) \leq S_6.$$

Consequently, as proved in Corollary 24.1,

$$\mathrm{Gal}(L_\zeta K^{\mathrm{sol}}/K^{\mathrm{sol}}) \cong A_6, \qquad F_\zeta \text{ is irreducible over } K^{\mathrm{sol}}.$$

The discriminant decides between the two possible groups. In Corollary 24.1 we prove, more precisely, that the splitting field meets K^{sol} in its sign subfield, whereas every degree-six root field meets K^{sol} only in K. This yields exact factorizations over arbitrary finite fields whose Galois closures over $\mathbf{Q}$ are solvable; see Theorems 25.1 and 26.1.

Theorem 1.4 (Abelian-stable irreducibility)**.** *Let $\zeta \neq 1$ be any root of unity and put $K = \mathbf{Q}(\zeta)$. Then*

$$F_\zeta(T) \text{ is irreducible over } K^{\mathrm{ab}}.$$

Equivalently, if α is a root of F_ζ, then

$$K(\alpha) \cap K^{\mathrm{ab}} = K.$$

The abelian theorem also admits an independent proof through an abstract sextic disjointness criterion. We retain that proof because the criterion does not depend on the special form of this family.

The same sextic family also exhibits full generic monodromy. We prove that its splitting field over $\mathbf{Q}(X)$ is a regular S_6-extension and that, outside finitely many roots of unity ζ, the specialized splitting field still has group S_6 over $\mathbf{Q}^{\mathrm{ab}}$. Over $\mathbf{Q}(\zeta)^{\mathrm{ab}}$ the corresponding group is then A_6; see Theorems 20.1 and 20.2.

The normalization of the multiplier curve transfers Theorem 1.2 to the period-4 delta factors.

Corollary 1.5. *For every integer $k \geq 1$, the period-4 delta factor $\Delta_{4k,4}$ is irreducible over* $\mathbf{Q}$.

Theorem 1.1 is proved in Sections 2–5. The proof of Theorem 1.2 combines reduction at primes above 2 with a uniform archimedean root bound. For Theorem 1.4, an abstract criterion combines an irreducible local quartic factor with the absence of a ground-field root of a $3+3$ resolvent. To prove Theorem 1.3, we construct a second sextic resolvent attached to the exceptional transitive subgroup $S_5 < S_6$. Its Newton polygon forces all of its factor degrees to be divisible by three. The two resolvents, together with the classification of primitive subgroups of S_6, leave only A_6 and S_6. All substantial explicit identities, congruences, discriminants, and finite permutation computations are accompanied by deterministic exact-integer certificates in Appendix A.

Part I
Cyclotomic specialization on unicritical multiplier curves

2 A Kummer–torsion specialization theorem

For a number field B, write

$$B^c = B(\mu_\infty).$$

Thus $\mathbf{Q}^c = \mathbf{Q}^{\mathrm{ab}}$ by Kronecker–Weber. We first record a one-dimensional form of cyclotomic Hilbert irreducibility that will be used twice below.

Let $X/\mathbf{Q}$ be a geometrically integral normal curve and let $t, y \in \mathbf{Q}(X)$ satisfy

$$\mathbf{Q}(X) = \mathbf{Q}(t, y), \qquad N = [\mathbf{Q}(X) : \mathbf{Q}(y)].$$

Regard y as a finite cover of $\mathbf{G}_m$ after deleting its zeros, poles, and finitely many other points. Let

$$A(T, Y) \in \mathbf{Q}(Y)[T]$$

be the monic minimal polynomial of t over $\mathbf{Q}(Y)$. There is a finite Galois-stable set $\Sigma_{\text{alg}} \subset \mu_\infty$ such that, for $\theta \notin \Sigma_{\text{alg}}$, specialization at $Y = \theta$ is defined, $A(T, \theta)$ is separable of degree N, and its roots are the t-coordinates of the points of $y^{-1}(\theta)$. Indeed, this follows by shrinking the target so that the cover is finite étale and the primitive-element plane model is isomorphic to it.

We say that t separates the torsion fibers of y away from a finite set if there is a finite Galois-stable $\Sigma_{\text{sep}} \subset \mu_\infty$ such that

$$P \longmapsto t(P) \tag{2}$$

is injective on

$$\bigcup_{\theta \in \mu_\infty \setminus \Sigma_{\text{sep}}} y^{-1}(\theta)(\overline{\mathbf{Q}}).$$

This single condition has two logically distinct consequences: fibers over different torsion points have disjoint sets of t-coordinates, and the torsion label can be recovered from any one such coordinate.

Recall that a cover $y : X \to \mathbf{G}_m$ satisfies the pullback condition *(PB)* if its pullback by $u \mapsto u^n$ is geometrically irreducible for every $n \geq 1$ [Zan10, Definition and Proposition 2.1]. Equivalently,

$$U^n - y \quad \text{is irreducible in } \overline{\mathbf{Q}}(X)[U] \quad (n \geq 1). \tag{3}$$

Theorem 2.1 (Kummer–torsion specialization)**.** *Assume that y satisfies (PB) and that t separates its torsion fibers away from a finite set. Fix $r \geq 1$ and put $x = y^r$. For $k \geq 2$, set*

$$S_r(k) = \{\theta \in \mu_\infty : \operatorname{ord}(\theta^r) = k\}$$

and, whenever all the specializations are good, put

$$\Psi_{r,k}(T) = \prod_{\theta \in S_r(k)} A(T, \theta). \tag{4}$$

Then the following assertions hold for every sufficiently large k.

(i) *Over $\mathbf{Q}^{\text{ab}}$, the polynomial $\Psi_{r,k}$ has exactly $r\varphi(k)$ distinct irreducible factors, namely the $A(T, \theta)$, and every factor has degree N.*

(ii) *The possible orders of elements of $S_r(k)$ are precisely*

$$n = ka, \qquad a \mid r, \qquad (k, r/a) = 1. \tag{5}$$

For every such a, the polynomial

$$F_{k,a}(T) = \prod_{\operatorname{ord}(\theta) = ka} A(T, \theta) \tag{6}$$

belongs to $\mathbf{Q}[T]$, is irreducible there, and has degree $N\varphi(ka)$. The product of these $F_{k,a}$ is the complete factorization of $\Psi_{r,k}$ over $\mathbf{Q}$.

(iii) *If t_0 is a root of $A(T, \theta)$, then*

$$\theta \in \mathbf{Q}(t_0), \qquad \mathbf{Q}(t_0) \cap \mathbf{Q}^{\text{ab}} = \mathbf{Q}(\theta). \tag{7}$$

Proof. By Zannier's cyclotomic Hilbert irreducibility theorem [Zan10, Theorem 2.1], condition (PB) gives a finite set $\Sigma_{\mathrm{Zan}} \subset \mu_\infty$ outside which the fiber of y is irreducible over $\mathbf{Q}^c = \mathbf{Q}^{\mathrm{ab}}$ and has degree N. Enlarge this set by Σ_{alg} and Σ_{sep}. Since $|S_r(k)| = r\varphi(k)$ and every $\theta \in S_r(k)$ has order at least k, no element of this fixed finite set occurs once k is large. This proves the irreducibility and degree assertion in (i); separation shows that the factors have disjoint root sets.

Let $P \in y^{-1}(\theta)$ and write $t_0 = t(P)$. If $\sigma \in \mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}(t_0))$, then

$$t(\sigma P) = t_0, \qquad y(\sigma P) = \sigma\theta.$$

Torsion separation forces $\sigma P = P$, and hence $\sigma\theta = \theta$. Thus $\theta \in \mathbf{Q}(t_0)$. Irreducibility over $\mathbf{Q}^{\mathrm{ab}}$ gives

$$[\mathbf{Q}^{\mathrm{ab}}(t_0) : \mathbf{Q}^{\mathrm{ab}}] = N.$$

It also implies that $A(T, \theta)$ is irreducible over $\mathbf{Q}(\theta)$, so $[\mathbf{Q}(t_0) : \mathbf{Q}(\theta)] = N$. If $D = \mathbf{Q}(t_0) \cap \mathbf{Q}^{\mathrm{ab}}$, the compositum degree formula gives $[\mathbf{Q}(t_0) : D] = N$. Since $\mathbf{Q}(\theta) \subseteq D$, the two degree equalities force $D = \mathbf{Q}(\theta)$, proving (iii).

If $n = \mathrm{ord}(\theta)$, then

$$\mathrm{ord}(\theta^r) = \frac{n}{(n, r)}.$$

Writing $a = (n, r)$ proves (5), and the converse is immediate. The product in (6) is Galois invariant. A root t_0 of one of its factors has, by (iii), degree

$$[\mathbf{Q}(t_0) : \mathbf{Q}] = N\varphi(ka),$$

which is the degree of $F_{k,a}$. Hence $F_{k,a}$ is the minimal polynomial of t_0. Distinct a give disjoint root sets, completing the proof. □

The arithmetic count in the preceding theorem is classical; compare the standard irreducibility criteria for cyclotomic compositions in [Dam11]. It is included to make the later dynamical factorization exact.

Corollary 2.2. *For all sufficiently large k, the number of irreducible factors of $\Psi_{r,k}$ over $\mathbf{Q}$ is*

$$\mathfrak{f}_r(k) = \prod_{\substack{p^\alpha \| r \\ p \nmid k}} (\alpha + 1). \tag{8}$$

In particular,

$$\Psi_{r,k} \text{ is irreducible over } \mathbf{Q} \iff \mathrm{rad}(r) \mid k. \tag{9}$$

Consequently the set of such k has natural density $1/\mathrm{rad}(r)$.

Proof. For a prime power $p^\alpha \| r$, condition $(k, r/a) = 1$ forces the exponent of p in a to equal α when $p \mid k$, while all $\alpha + 1$ exponents are allowed when $p \nmid k$. This proves (8) and (9). The density statement is unchanged by the finite exceptional set. □

We shall also need the same result for a cover that is itself defined only over a number field. The finite exceptional set may depend on that field.

Theorem 2.3 (Number-field Kummer–torsion specialization)**.** *Let B be a number field, let Z/B be a geometrically integral normal curve, and let $u, v \in B(Z)$ satisfy*

$$B(Z) = B(u, v), \qquad N = [B(Z) : B(v)].$$

Let $A_B(T,Y) \in B(Y)[T]$ be the monic minimal polynomial of u over $B(Y)$. Assume that $v : Z \dashrightarrow \mathbf{G}_m$ satisfies (PB) and that u separates the torsion fibers of v away from a finite Galois-stable set. Fix $r \geq 1$, put

$$S_r(k) = \{\theta \in \mu_\infty : \mathrm{ord}(\theta^r) = k\}, \qquad \Psi^B_{r,k}(T) = \prod_{\theta \in S_r(k)} A_B(T,\theta),$$

whenever all specializations are defined. Then the following hold for every sufficiently large k.

(i) *Over $B^c = B(\mu_\infty)$, the distinct irreducible factors of $\Psi^B_{r,k}$ are precisely the $A_B(T,\theta)$, each of degree N.*

(ii) *Over B, the irreducible factors are indexed by the* $\mathrm{Gal}(B^c/B)$*-orbits $\mathcal{O}$ on $S_r(k)$. The orbit $\mathcal{O}$ gives a factor of degree $N|\mathcal{O}|$.*

(iii) *If u_0 is a root of $A_B(T,\theta)$, then*

$$\theta \in B(u_0), \qquad B(u_0) \cap B^c = B(\theta). \tag{10}$$

For an admissible order $n = ka$, where $a \mid r$ and $(k, r/a) = 1$, put $K_n = \mathbf{Q}(\zeta_n)$ and $D_n = B \cap K_n$. The order-n block has $[D_n : \mathbf{Q}]$ irreducible factors over B, each of degree $N[K_n : D_n]$.

Proof. Apply Zannier's theorem [Zan10, Theorem 2.1] over B. In dimension one its exceptional union of proper torsion cosets is a finite set of torsion points. Enlarge it by the finite sets on which specialization or separation fails. Thus, for every remaining torsion point θ, the fiber $v^{-1}(\theta)$ is irreducible over B^c and has degree N. Every $\theta \in S_r(k)$ has order at least k, so all labels are good once k is sufficiently large; separation makes the resulting factors pairwise distinct. This proves (i).

Let $P \in v^{-1}(\theta)$ and put $u_0 = u(P)$. If $\sigma \in \mathrm{Gal}(\overline{\mathbf{Q}}/B(u_0))$, then

$$u(\sigma P) = u_0, \qquad v(\sigma P) = \sigma\theta.$$

Separation gives $\sigma P = P$, hence $\sigma\theta = \theta$ and $\theta \in B(u_0)$. Irreducibility over B^c and over the subfield $B(\theta)$ gives

$$[B^c(u_0) : B^c] = N = [B(u_0) : B(\theta)].$$

For $D = B(u_0) \cap B^c$, the compositum degree formula gives $[B(u_0) : D] = N$. Since $B(\theta) \subseteq D$, the two degree equalities force $D = B(\theta)$, proving (iii).

For a $\mathrm{Gal}(B^c/B)$-orbit $\mathcal{O}$, the product of the factors labelled by $\mathcal{O}$ belongs to $B[T]$. If u_0 lies above $\theta \in \mathcal{O}$, then (iii) gives

$$[B(u_0) : B] = N[B(\theta) : B] = N|\mathcal{O}|,$$

which is the degree of that product. It is therefore the minimal polynomial of u_0 over B. This proves (ii). Finally, restriction identifies the action on primitive n-th roots with $\mathrm{Gal}(K_n/D_n)$. This action is free, so it has $[D_n : \mathbf{Q}]$ orbits, each of size $[K_n : D_n]$, proving the last assertion. □

3 Unicritical multiplier curves and the reduced multiplier

Fix integers $d \geq 2$ and $m \geq 1$, and put

$$f_c(z) = z^d + c, \qquad e = d - 1, \qquad \nu_d(m) = \sum_{r|m} \mu(m/r) d^r.$$

The m-th multiplier polynomial $M_{d,m}(c, x)$ is the monic polynomial whose roots, for generic c, are the multipliers of the cycles of exact period m. It has degree $\nu_d(m)/m$ in x. We use the following standard facts about this polynomial and the corresponding dynatomic curve:

(a) the marked dynatomic curve is geometrically irreducible;

(b) $M_{d,m}$ is geometrically irreducible;

(c) there is a polynomial $D_{d,m}(t, x) \in \mathbf{Z}[t, x]$ such that

$$M_{d,m}(c, x) = D_{d,m}(d^d c^e, x), \tag{11}$$

and $D_{d,m}$ has degree $\nu_d(m)/d$ in t, with leading coefficient ± 1.

The geometric irreducibility statements go back to Morton [Mor96, Corollary 1]; the integrality, degree, and leading coefficient in the scaled parameter are proved in [MST25, Theorem 1.2 and Propositions 3.3, 3.5]. See also [Hug21, Theorem 1.38 and Propositions 1.51, 1.59].

Let $\mathcal{Y}_{d,m}$ be the normalization of the marked dynatomic curve, and write

$$L = \overline{\mathbf{Q}}(\mathcal{Y}_{d,m}).$$

Iteration induces an automorphism σ of order m. If z is the marked point, put

$$z_j = f_c^j(z), \qquad P = \prod_{j=0}^{m-1} z_j. \tag{12}$$

Then $P \in L^{\langle\sigma\rangle}$ and the cycle multiplier is

$$x = (f_c^m)'(z) = d^m P^e. \tag{13}$$

Lemma 3.1 (The cycle and scaling quotients)**.** *One has*

$$L^{\langle\sigma\rangle} = \overline{\mathbf{Q}}(c, x). \tag{14}$$

The action

$$\xi \cdot (c, z_0, \ldots, z_{m-1}) = (\xi c, \xi z_0, \ldots, \xi z_{m-1}), \qquad \xi \in \mu_e, \tag{15}$$

commutes with σ, fixes x, and has fixed field

$$\overline{\mathbf{Q}}(c, x)^{\mu_e} = \overline{\mathbf{Q}}(c^e, x) = \overline{\mathbf{Q}}(t, x), \qquad t = d^d c^e. \tag{16}$$

Proof. The inclusion $\overline{\mathbf{Q}}(c, x) \subseteq L^{\langle\sigma\rangle}$ follows from (13). The two extensions over $\overline{\mathbf{Q}}(c)$ have the same degree:

$$[L^{\langle\sigma\rangle} : \overline{\mathbf{Q}}(c)] = \frac{\nu_d(m)}{m} = \deg_x M_{d,m}.$$

Here geometric irreducibility of $M_{d,m}$ is essential: it identifies $\overline{\mathbf{Q}}(c, x)$ with the function field of the multiplier curve and prevents a hidden proper subfield. This proves (14).

Since $\xi^d = \xi$, one has $f_{\xi c}(\xi z) = \xi f_c(z)$, proving that (15) is an action. It sends P to $\xi^m P$ and hence fixes x. Its action on the transcendental element c is faithful, so the fixed field is $\overline{\mathbf{Q}}(c^e, x)$, which is also $\overline{\mathbf{Q}}(t, x)$. $\square$

Let $\mathcal{X}_{d,m}$ denote the normalization of the projective closure of $D_{d,m}(t,x) = 0$. Thus

$$\mathbf{Q}(\mathcal{X}_{d,m}) = \mathbf{Q}(t,x).$$

Set

$$\gamma = (m,e), \qquad q = e/\gamma. \tag{17}$$

Proposition 3.2 (The reduced multiplier)**.** *The function*

$$y = d^{m/\gamma} P^q \tag{18}$$

belongs to $\mathbf{Q}(t,x)$ *and satisfies*

$$x = y^\gamma. \tag{19}$$

Moreover,

$$\deg(y : \mathcal{X}_{d,m} \longrightarrow \mathbf{P}^1) = N_{d,m} := \frac{\nu_d(m)}{d\gamma}. \tag{20}$$

Proof. Because mq is divisible by e, the expression in (18) is fixed by μ_e. It is defined over $\mathbf{Q}$, so Galois descent and Lemma 3.1 give $y \in \mathbf{Q}(t,x)$. Equation (19) follows at once from (13). Finally, the degree of the map x on $\mathcal{X}_{d,m}$ is $\deg_t D_{d,m} = \nu_d(m)/d$. Since $x = y^\gamma$, degrees of finite maps give (20). □

The next theorem is the geometric core of the generalization. The stabilizer h appearing in its proof cannot in general be omitted.

Theorem 3.3 (Maximal isogeny and pullback irreducibility)**.** *The reduced multiplier*

$$y : \mathcal{X}_{d,m} \dashrightarrow \mathbf{G}_m$$

satisfies (PB). Furthermore, $\gamma = (m, d-1)$ *is the largest geometric isogeny degree of the multiplier map* x*: if*

$$x = w^r \qquad \textit{in } \overline{\mathbf{Q}}(\mathcal{X}_{d,m})$$

for some $r \geq 1$*, then* $r \mid \gamma$*.*

Proof. We first suppose $m > 1$. Write $\Phi^*_{d,m}(z,c)$ for the m-th dynatomic polynomial of f_c. We first verify directly that its critical specialization is squarefree. Put

$$a_n(c) = f_c^{\circ n}(0) \qquad (n \geq 1).$$

Then $a_1 = c$ and, for $n \geq 2$, one has $a_n = a_{n-1}^d + c$, so $a_n \in \mathbf{Z}[c]$ is monic and

$$a_n' = d a_{n-1}^{d-1} a_{n-1}' + 1.$$

If β were a common root of a_n and a_n', then β would be an algebraic integer, as would $a_{n-1}(\beta)^{d-1} a_{n-1}'(\beta)$. The displayed identity would give

$$a_{n-1}(\beta)^{d-1} a_{n-1}'(\beta) = -\frac{1}{d},$$

which is impossible because $d \geq 2$ and the rational algebraic integers are the integers. Thus every a_n is squarefree. The dynatomic product identity, specialized at $z = 0$, is

$$a_n(c) = \prod_{r \mid n} \Phi^*_{d,r}(0,c).$$

Consequently every factor $\Phi^*_{d,r}(0,c)$ is squarefree. Moreover, Möbius inversion applied to degrees gives

$$\deg_c \Phi^*_{d,m}(0,c) = \sum_{r|m} \mu(m/r) d^{r-1} = \frac{\nu_d(m)}{d} > 0.$$

Compare [Hug21, Proposition 1.58]. Choose a root c_0 of $\Phi^*_{d,m}(0,c)$. Because the multiplier of the critical orbit is zero, the formal-period criterion shows that 0 has exact period m for f_{c_0}; in particular, $c_0 \neq 0$. Squarefreeness gives

$$\partial_c \Phi^*_{d,m}(0,c_0) \neq 0.$$

Thus the marked dynatomic curve is smooth at $(0,c_0)$ and $z = z_0$ is a local parameter there. The remaining points $z_1, \dots, z_{m-1}$ of the critical cycle are nonzero, so $P = z_0 \cdots z_{m-1}$ has a simple zero. Exact periodicity makes the cycle-rotation stabilizer trivial, and the scaling action is free because $c_0 \neq 0$. Consequently both quotient maps are unramified at this point. At its image P_0 on $\mathcal{X}_{d,m}$ one therefore has

$$\operatorname{ord}_{P_0}(y) = q, \qquad \operatorname{ord}_{P_0}(x) = e. \tag{21}$$

We next compute an order at infinity. Let $\overline{\mathcal{Y}}_{d,m}$ be the smooth projective model of the marked dynatomic curve, choose a point R_∞ above $c = \infty$, and normalize the base change given by $c = s^{-d}$. Choose a point $\widetilde{R}_\infty$ above $(R_\infty, s = 0)$, write $v = \operatorname{ord}_{\widetilde{R}_\infty}$, and put

$$b_j = \frac{v(z_j)}{v(s)}.$$

The recurrence $z_{j+1} = z_j^d + c$ forces $b_j = -1$ for every j. Indeed, if the minimum of the b_j were less than -1, then at an index where it is attained the term z_j^d would have strictly smaller valuation than c, giving $b_{j+1} = db_j < b_j$, a contradiction. Thus all $b_j \geq -1$. If some $b_j > -1$, then c would be the unique term of least valuation and would give $b_{j+1} = -d < -1$, another contradiction.

It follows that $z_j = s^{-1} u_j$ with every u_j a unit. The orbit equations become

$$u_j^d + 1 = s^e u_{j+1} \qquad (j \bmod m). \tag{22}$$

Write $a_j = u_j(0)$, so $a_j^d = -1$. At $s = 0$ the Jacobian with respect to $u_0, \dots, u_{m-1}$ is the invertible diagonal matrix

$$\operatorname{diag}(da_0^{d-1}, \dots, da_{m-1}^{d-1}).$$

The formal implicit-function theorem therefore identifies the completed local ring at $\widetilde{R}_\infty$ with $\overline{\mathbf{Q}}[[s]]$. Uniqueness of the solution, and the fact that the equations depend on s only through s^e, also give

$$u_j(s) \in \overline{\mathbf{Q}}[[s^e]]^\times.$$

The deck group μ_d of the base change acts by

$$(s, u_0, \dots, u_{m-1}) \longmapsto (\omega s, \omega u_0, \dots, \omega u_{m-1}), \qquad \omega \in \mu_d.$$

It leaves the original functions c and $z_j = s^{-1}u_j$ unchanged. At $s = 0$ it sends $(a_j)_j$ to $(\omega a_j)_j$, so no nonidentity element fixes $\widetilde{R}_\infty$. The quotient of the normalized base change by this deck action is $\overline{\mathcal{Y}}_{d,m}$; hence the map back to the marked curve is unramified at $\widetilde{R}_\infty$. From

$$c = s^{-d}, \qquad P = s^{-m} \prod_{j=0}^{m-1} u_j(s)$$

we obtain on the marked curve

$$\operatorname{ord}_{R_\infty}(c) = -d, \qquad \operatorname{ord}_{R_\infty}(P) = -m. \tag{23}$$

Let h be the order of the stabilizer of R_∞ under cycle rotation. In characteristic zero the marked-to-cycle quotient has ramification index h at this point. Since both c and P descend, at the image Q_∞ one has

$$\operatorname{ord}_{Q_\infty}(c) = -\frac{d}{h}, \qquad \operatorname{ord}_{Q_\infty}(P) = -\frac{m}{h}. \tag{24}$$

In particular, $h \mid d$ and $h \mid m$.

Every element of μ_e fixes Q_∞. On the normalized base change the scaling action by $\xi \in \mu_e$ lifts locally as

$$(s, u_0, \ldots, u_{m-1}) \longmapsto (\xi^{-1}s, u_0, \ldots, u_{m-1}).$$

This covers $(c, z_j) \mapsto (\xi c, \xi z_j)$ and preserves the chosen branch because $u_j(s) \in \overline{\mathbf{Q}}[[s^e]]$. Thus the induced action on the cycle quotient is faithful, since ξ sends the nonconstant function c to ξc. Hence the stabilizer of Q_∞ is all of μ_e, and in characteristic zero the cycle-to-scaling quotient has ramification index exactly e at Q_∞. Since $y = d^{m/\gamma}P^q$ and $x = d^m P^e$, their orders at the image P_∞ on $\mathcal{X}_{d,m}$ are

$$\operatorname{ord}_{P_\infty}(y) = -\frac{m}{\gamma h}, \qquad \operatorname{ord}_{P_\infty}(x) = -\frac{m}{h}. \tag{25}$$

Write $m = \gamma r$. Since $e = \gamma q$ and $(q, r) = 1$, while $h \mid d$ and $\gamma \mid d-1$, one has $(h, \gamma) = 1$. The divisibility $h \mid m = \gamma r$ therefore gives $h \mid r$, and hence

$$\gcd\left(q, \frac{m}{\gamma h}\right) = \gcd\left(q, \frac{r}{h}\right) = 1. \tag{26}$$

The two orders in (21) and (25) show that y is not a p-th power in $\overline{\mathbf{Q}}(\mathcal{X}_{d,m})$ for any prime p. The binomial criterion [Lan02, Chapter VI, §9, Theorem 9.1] now makes $U^n - y$ irreducible for every $n \geq 1$. When $4 \mid n$, the exceptional condition $y \in -4\overline{\mathbf{Q}}(\mathcal{X}_{d,m})^4$ is already excluded because $-4 = (1+i)^4$ in the constant field. This proves (PB).

The two displayed orders of x have greatest common divisor

$$\gcd\left(e, \frac{m}{h}\right) = \gamma \gcd\left(q, \frac{r}{h}\right) = \gamma.$$

If $x = w^\ell$, every divisor order of x is divisible by ℓ, so $\ell \mid \gamma$. Since $x = y^\gamma$, the bound is attained.

It remains to treat $m = 1$. If z is a fixed point, then $c = z - z^d$ and $x = dz^{d-1}$. Direct elimination gives

$$t = x(d-x)^{d-1}. \tag{27}$$

Here $\gamma = 1$ and $y = x$; in particular y has order one at the center. It satisfies (PB), and the maximal isogeny degree is 1. Notice that the center is fully ramified in the scaling quotient in this case, which is why it was separated from the preceding argument. □

4 Torsion separation on the unicritical quotient

To apply Theorem 2.1, it remains to recover the reduced multiplier from the scaled parameter at torsion fibers.

Proposition 4.1 (Torsion separation)**.** *On the union of the finite fibers*

$$y^{-1}(\theta), \qquad \theta \in \mu_\infty, \quad \theta^\gamma \neq 1,$$

the function $t = d^d c^{d-1}$ is injective. In particular, if $Q \in y^{-1}(\theta)$ and $t_0 = t(Q)$, then

$$\theta \in \mathbf{Q}(t_0).$$

Proof. First, a formal-period-m point with $x \neq 1$ has exact period m. Indeed, if its exact period were $r < m$, the formal-period criterion [Sil07, Theorem 4.5(b)] would write $m = rs$, with the r-cycle multiplier a primitive s-th root of unity; its multiplier for the m-th iterate would then be 1.

A unicritical polynomial has at most one finite cycle whose multiplier is a root of unity. Indeed, after passing to a suitable iterate, such a cycle is parabolic with multiplier 1. The Fatou critical-point theorem for parabolic basins [Mil06, Theorem 10.15] supplies a critical point of that iterate whose orbit tends to the cycle. By the chain rule, one of its forward images is a critical point of the original polynomial, so the cycle attracts a critical orbit of f_c. The resulting critical point of f_c is finite, since infinity lies in the basin of the superattracting fixed point at infinity. The only finite critical point of $z^d + c$ is 0, and the attracting basins of two distinct periodic cycles are disjoint, so two such cycles cannot coexist. Also $c = 0$ cannot occur here: for z^d, a nonzero periodic orbit has multiplier a positive power of d, while the zero fixed point has multiplier zero.

Suppose two points Q_1, Q_2 in the indicated torsion fibers have the same t-coordinate, represented by parameters c_1, c_2. Since these parameters are nonzero, there is a unique $\xi \in \mu_e$ with $c_2 = \xi c_1$. The conjugacy $z \mapsto \xi z$ transports the first cycle to a cycle for f_{c_2}. The transported cycle and the second cycle coincide by the preceding paragraph. Moreover $P \mapsto \xi^m P$, whereas

$$\xi^{mq} = 1,$$

so the value of $y = d^{m/\gamma} P^q$ is unchanged. Thus $Q_1 = Q_2$.

There is no hidden splitting on normalization: at these points $\partial_z(f_c^m(z) - z) = x - 1 \neq 0$, cycle rotation is free, and the μ_e-action is free because $c \neq 0$. Both quotient maps are étale there. The final assertion now follows either directly or from the Galois argument used in Theorem 2.1. □

5 Eventual exact factorization in the scaled parameter

Let $A_{d,m}(T, Y)$ be the monic minimal polynomial of t over $\mathbf{Q}(y)$. It has degree $N_{d,m}$ in T. Since $D_{d,m}(T, Y^\gamma)$ is monic up to a fixed sign in T, Gauss's lemma and normality of $\mathbf{Q}[Y]$ allow $A_{d,m}$ to be normalized in $\mathbf{Q}[Y, T]$. Over $\overline{\mathbf{Q}}$ one has

$$D_{d,m}(T, Y^\gamma) \doteq \prod_{\xi \in \mu_\gamma} A_{d,m}(T, \xi Y), \tag{28}$$

where $\doteq$ denotes equality up to the fixed leading sign. Each $A_{d,m}(T, \xi Y)$ is irreducible and divides the left-hand side. The factors are distinct: otherwise $A_{d,m}(T, Y) = A_{d,m}(T, \delta Y)$

for some nontrivial $\delta \in \mu_\gamma$. Then $t \mapsto t$ and $y \mapsto \delta y$ would induce a nontrivial automorphism of $\overline{\mathbf{Q}}(t, y) = \overline{\mathbf{Q}}(\mathcal{X}_{d,m})$ fixing both t and $x = y^\gamma$. This contradicts $\overline{\mathbf{Q}}(\mathcal{X}_{d,m}) = \overline{\mathbf{Q}}(t, x)$. The factors are therefore pairwise coprime, and comparison of degrees in T gives (28).

For $k \geq 2$, define the scaled unicritical delta polynomial by taking the positive-leading normalization of

$$\mathcal{P}_{d,m,k}(T) \doteq \mathrm{Res}_X\big(\Phi_k(X), D_{d,m}(T, X)\big). \tag{29}$$

In the standard unicritical notation of [MST25, Theorem 1.1],

$$\mathcal{P}_{d,m,k}(d^d c^{d-1}) \doteq \Delta_{mk,m}(c),$$

where $\doteq$ accounts only for the chosen leading-sign normalization. By (28),

$$\mathcal{P}_{d,m,k}(T) = \prod_{\mathrm{ord}(\theta^\gamma)=k} A_{d,m}(T, \theta). \tag{30}$$

Theorem 5.1 (Eventual exact factorization)**.** *Fix $d \geq 2$ and $m \geq 1$, and retain*

$$\gamma = (m, d-1), \qquad N_{d,m} = \frac{\nu_d(m)}{d\gamma}.$$

For all sufficiently large $k \geq 2$:

(i) *over $\mathbf{Q}^{\mathrm{ab}}$, $\mathcal{P}_{d,m,k}$ has exactly $\gamma\varphi(k)$ distinct irreducible factors, each of degree $N_{d,m}$;*

(ii) *over $\mathbf{Q}$, its complete irreducible factorization is indexed by*

$$a \mid \gamma, \qquad (k, \gamma/a) = 1,$$

and the factor indexed by a has degree $N_{d,m}\varphi(ka)$;

(iii) *if t_0 belongs to the factor indexed by a reduced multiplier θ of order ka, then*

$$\mathbf{Q}(t_0) \cap \mathbf{Q}^{\mathrm{ab}} = \mathbf{Q}(\theta). \tag{31}$$

The corresponding factorization over any fixed number field B is given by Theorem 2.3.

Proof. Theorem 3.3 supplies (PB), and Proposition 4.1 supplies torsion separation for every $x \neq 1$. Since $k \geq 2$, all fibers in (30) have $x \neq 1$. Apply Theorem 2.1. □

Corollary 5.2. *For all sufficiently large k,*

$$\mathcal{P}_{d,m,k} \text{ is irreducible over } \mathbf{Q} \iff \mathrm{rad}((m, d-1)) \mid k. \tag{32}$$

If $(m, d-1) = 1$, the whole row is therefore eventually irreducible. For the quadratic family $d = 2$, this holds for every fixed m. Thus every fixed row of the Morton–Vivaldi conjecture has only finitely many possible exceptions.

Proof. This is Corollary 2.2. When $d = 2$, the scaled parameter $t = 4c$ is an invertible linear change of the original parameter, so no extra composition issue occurs. □

Remark 5.3 (Effectivity)**.** *Zannier notes that the arguments used for the toric theorem are effective [Zan10, p. 400]. Hence, for each fixed pair (d, m) and fixed base number field, the finite exceptional set is effectively computable in principle. No practical numerical bound is claimed here.*

6 Descent to the original parameter

The scaled coordinate $t = d^d c^{d-1}$ is the natural coordinate on the unicritical moduli quotient. Returning to the original c-line introduces a second, genuine Kummer descent. We record the resulting factorization, both to delimit the role of the scaled coordinate and to show that this descent can still be made exact.

Retain

$$e = d - 1, \qquad \gamma = (m, e), \qquad q = e/\gamma, \qquad a = m/\gamma,$$

so $(a, q) = 1$. Choose $s \in \overline{\mathbf{Q}}$ with $s^q = d$ and put

$$B_0 = \mathbf{Q}(s), \qquad \rho = s^a P. \tag{33}$$

Let $\mathcal{C}_{d,m}/\mathbf{Q}$ be the normalization of the projective multiplier curve $M_{d,m}(c, x) = 0$; thus $\mathbf{Q}(\mathcal{C}_{d,m}) = \mathbf{Q}(c, x)$. By Lemma 3.1 and Galois descent, the cycle product P belongs to $\mathbf{Q}(c, x)$ and satisfies $x = d^m P^e$. Since $ae = mq$, one has

$$\rho^e = s^{ae} P^e = d^m P^e = x.$$

Moreover,

$$B_0(c, x) = B_0(c, P) = B_0(c, \rho). \tag{34}$$

Indeed, the first equality follows in both directions from $P \in \mathbf{Q}(c, x)$ and $x = d^m P^e$, and the second is immediate from $\rho = s^a P$ with $s \in B_0$. The coprimality $(a, q) = 1$ instead records that B_0 is already generated by the constant used in ρ: if $au + qv = 1$, then $s = (s^a)^u d^v$, so $\mathbf{Q}(s^a) = \mathbf{Q}(s)$.

Proposition 6.1 (The original-parameter PB cover)**.** *On the cycle quotient in the original parameter c, the map*

$$\rho : \mathcal{C}_{d,m} \dashrightarrow \mathbf{G}_m$$

has degree

$$L_{d,m} = \frac{\nu_d(m)}{d} \tag{35}$$

and satisfies (PB) over B_0. Moreover, c separates all torsion fibers with $x = \rho^e \neq 1$.

Proof. The function P, and hence ρ, has a simple zero at every exact period-m critical center. Therefore ρ is not a p-th power for any prime p in the geometric function field. The binomial criterion, including its harmless fourth-power clause as in the proof of Theorem 3.3, proves (PB).

The scaled-parameter degree formula gives

$$[B_0(c, x) : B_0(x)] = \deg_c M_{d,m} = \frac{e\nu_d(m)}{d}.$$

Because ρ is transcendental and $x = \rho^e$, one has $[B_0(\rho) : B_0(x)] = e$. The tower

$$B_0(x) \subset B_0(\rho) \subset B_0(c, \rho) = B_0(c, x)$$

therefore gives $[B_0(c, \rho) : B_0(\rho)] = \nu_d(m)/d$, proving (35). Finally, if the same c occurred in two torsion fibers, it would give two distinct finite root-of-unity multiplier cycles for one unicritical polynomial. The uniqueness argument in Proposition 4.1 excludes this, and the quotient is étale at $x \neq 1$. $\square$

Let $H_{d,m}(C,R)$ be a normalized primitive-element equation for c over $B_0(\rho)$; it has degree $L_{d,m}$ in C. Up to a nonzero constant independent of C,

$$M_{d,m}(C,R^e) \doteq \prod_{\xi\in\mu_e} H_{d,m}(C,\xi R). \tag{36}$$

Indeed, after adjoining R with $R^e = x$, the base change decomposes into the e components $\rho = \xi R$. Each is geometrically integral and has primitive-element equation $H_{d,m}(C,\xi R)$. These equations are distinct: an equality $H_{d,m}(C,R) = H_{d,m}(C,\delta R)$ with $1 \neq \delta \in \mu_e$ would induce a nontrivial automorphism $c \mapsto c$, $\rho \mapsto \delta\rho$ of $\overline{\mathbf{Q}}(c,\rho) = \overline{\mathbf{Q}}(c,x)$ fixing c and $x = \rho^e$, which is impossible. Comparison of degrees in C now gives (36). Define

$$\mathcal{R}_{d,m,k}(C) = \mathrm{Res}_X\big(\Phi_k(X), M_{d,m}(C,X)\big). \tag{37}$$

This is, up to the conventional normalization, the original unicritical delta polynomial, and

$$\mathcal{R}_{d,m,k}(C) \doteq \mathcal{P}_{d,m,k}(d^dC^e).$$

Theorem 6.2 (Original-parameter factorization)**.** *Fix d and m. For every sufficiently large $k \geq 2$ the following statements hold.*

(i) *Over $B_0^c = B_0(\mu_\infty)$, the polynomial $\mathcal{R}_{d,m,k}$ has exactly $e\varphi(k)$ distinct irreducible factors, each of degree $L_{d,m}$.*

(ii) *Put $K_n = \mathbf{Q}(\zeta_n)$ and $D_n = B_0 \cap K_n$. For every*

$$n = kb, \qquad b \mid e, \qquad (k, e/b) = 1,$$

the order-n block has $[D_n : \mathbf{Q}]$ irreducible factors over B_0, each of degree $L_{d,m}[K_n : D_n]$.

(iii) *If c_0 lies in the torsion fiber $\rho = \theta$, then*

$$B_0(c_0) \cap B_0^c = B_0(\theta). \tag{38}$$

(iv) *If $s \in \mathbf{Q}$—in particular, if $d-1 \mid m$—then the complete factorization over $\mathbf{Q}$ is indexed by*

$$b \mid e, \qquad (k, e/b) = 1,$$

and the factor indexed by b has degree $L_{d,m}\varphi(kb)$. Consequently,

$$\mathcal{R}_{d,m,k} \text{ is irreducible over } \mathbf{Q} \iff \mathrm{rad}(d-1) \mid k \tag{39}$$

for all sufficiently large k.

Proof. Apply Theorem 2.3 over B_0 to the PB cover ρ, with separating coordinate c and Kummer exponent e. This proves (i), (iii), and the orbit statement underlying (ii). Primitive n-th roots form $[D_n : \mathbf{Q}]$ orbits under $\mathrm{Gal}(B_0^c/B_0)$, each of size $[K_n : D_n]$, which proves (ii). If $s \in \mathbf{Q}$, the cover and its separating coordinate are defined over $\mathbf{Q}$; Theorem 2.1 and Corollary 2.2, with e in place of r, give (iv). □

There is also a complete descent to $\mathbf{Q}$ when $s \notin \mathbf{Q}$, but its correct indexing is by twisted Kummer orbits rather than by orders alone. This distinction prevents a false extension of (39).

Proposition 6.3 (Twisted orbit descent)**.** *Put $\Omega = B_0^c$. Then $\Omega/\mathbf{Q}$ is Galois. For $\sigma \in \mathrm{Gal}(\Omega/\mathbf{Q})$ set*

$$\kappa_\sigma = \frac{\sigma(s)}{s} \in \mu_q.$$

On

$$S_e(k) = \{\theta \in \mu_\infty : \mathrm{ord}(\theta^e) = k\}$$

define

$$\sigma * \theta = \kappa_\sigma^{-a}\sigma(\theta). \tag{40}$$

For all sufficiently large k, the irreducible factors of $\mathcal{R}_{d,m,k}$ over $\mathbf{Q}$ are indexed by the orbits $\mathcal{O}$ of this action, and the corresponding factor has degree

$$L_{d,m}|\mathcal{O}|. \tag{41}$$

Proof. The field Ω contains all conjugates ξs with $\xi \in \mu_q$, so it is normal over $\mathbf{Q}$. The quantities κ_σ form the usual Kummer cocycle:

$$\kappa_{\sigma\tau} = \kappa_\sigma\, \sigma(\kappa_\tau).$$

Thus (40) is a group action. Since $q \mid e$,

$$(\sigma * \theta)^e = \sigma(\theta^e),$$

so the action preserves $S_e(k)$.

The irreducible Ω-factors are the fibers $\rho = \theta$. If a point P_0 satisfies $\rho(P_0) = \theta$, then

$$\rho(\sigma P_0) = s^a\sigma(P(P_0)) = \kappa_\sigma^{-a}\sigma\big(s^a P(P_0)\big) = \kappa_\sigma^{-a}\sigma(\theta).$$

Thus $\mathrm{Gal}(\Omega/\mathbf{Q})$ permutes the distinct irreducible Ω-factors exactly by (40). Galois descent says that the product over one orbit belongs to $\mathbf{Q}[C]$. It is irreducible: after extension to Ω, a proper factor over $\mathbf{Q}$ would correspond to a nonempty proper $\mathrm{Gal}(\Omega/\mathbf{Q})$-stable subset of that orbit. Every $\mathbf{Q}$-factor arises in this way. Its degree is (41). □

Remark 6.4. *The twisted action is genuinely necessary. For $d = 3$, $m = 1$, and a primitive cubic multiplier, direct elimination gives*

$$729C^4 + 27C^2 + 169.$$

Over $\mathbf{Q}(\sqrt{3})$ its two torsion-order blocks are conjugate and merge over $\mathbf{Q}$. Thus the simple order indexing valid when $s \in \mathbf{Q}$ cannot be asserted in general.

Part II

The complete quadratic period-4 row

7 The fixed-degree reformulation

Let ζ be a root of unity different from 1, and recall

$$F_\zeta(T) = T^6 + 2T^5 + 4T^4 + 6T^3 + (\zeta - 5)T^2 - 8T - 16. \tag{42}$$

A direct calculation gives

$$g(T) - \zeta = -\frac{F_\zeta(T)}{T^2}. \tag{43}$$

Proposition 7.1. *Let $k \geq 2$, fix a primitive k-th root ζ_0, and put $K = \mathbf{Q}(\zeta_0)$. The following conditions are equivalent:*

(A) Γ_k *is irreducible over* $\mathbf{Q}$*;*

(B) F_{ζ_0} *is irreducible over* K*;*

(C) $F_{\zeta'}$ *is irreducible over* K *for every primitive k-th root* ζ'*.*

Proof. The cyclotomic factorization and (43) give

$$\Gamma_k(T) = (-1)^{\varphi(k)} T^{2\varphi(k)} \prod_{\substack{\zeta^k=1 \\ \zeta \text{ primitive}}} (g(T) - \zeta) = \prod_{\substack{\zeta^k=1 \\ \zeta \text{ primitive}}} F_\zeta(T).$$

Thus Γ_k is monic of degree $6\varphi(k)$. Its coefficients are algebraic integers fixed by the Galois group of $K/\mathbf{Q}$, so they belong to $\mathbf{Z}$.

Let α be a root of F_{ζ_0}. Since $F_{\zeta_0}(0) = -16$, we have $\alpha \neq 0$, and (43) yields

$$\zeta_0 = g(\alpha) \in \mathbf{Q}(\alpha).$$

Consequently $K \subseteq \mathbf{Q}(\alpha)$. If F_{ζ_0} is irreducible over K, then

$$[\mathbf{Q}(\alpha) : \mathbf{Q}] = [K(\alpha) : K][K : \mathbf{Q}] = 6\varphi(k).$$

The minimal polynomial of α over $\mathbf{Q}$ therefore has the same degree as Γ_k and divides it, so it equals Γ_k.

Conversely, suppose that Γ_k is irreducible. Then every root α of F_{ζ_0} has degree $6\varphi(k)$ over $\mathbf{Q}$. Since $K \subseteq \mathbf{Q}(\alpha)$, it follows that $[K(\alpha) : K] = 6$. The minimal polynomial of α over K has degree 6 and divides F_{ζ_0}, so F_{ζ_0} is irreducible. Finally, the automorphisms of $K/\mathbf{Q}$ act transitively on the primitive k-th roots and send F_{ζ_0} to the corresponding polynomials $F_{\zeta'}$. This proves all three equivalences. $\square$

At $\zeta = 1$ one has the exceptional factorization

$$F_1(T) = (T+2)(T^2+4)(T^3-2). \tag{44}$$

Set

$$P(T) = F_1(T).$$

Then

$$F_\zeta(T) = P(T) + (\zeta - 1)T^2. \tag{45}$$

8 A uniform archimedean bound

Lemma 8.1. *Let $|\zeta| = 1$. Every complex root α of F_ζ satisfies*

$$1 < |\alpha| < \frac{5}{2}.$$

Proof. Put $r = |\alpha|$. By (45),

$$|P(\alpha)| = |\zeta - 1| r^2 \leq 2r^2. \tag{46}$$

If $r \leq 1$, then the reverse triangle inequality and (44) give

$$|P(\alpha)| \geq (2 - r)(4 - r^2)(2 - r^3) \geq 3,$$

whereas $2r^2 \leq 2$, contradicting (46).

Now suppose that $r \geq 5/2$. Again by the reverse triangle inequality,

$$\frac{|P(\alpha)|}{r^2} \geq (r - 2)(r^2 - 4)\left(r - \frac{2}{r^2}\right).$$

Each of the three factors on the right is positive and increasing for $r \geq 5/2$. Hence

$$\frac{|P(\alpha)|}{r^2} \geq \frac{1}{2} \cdot \frac{9}{4} \cdot \frac{109}{50} = \frac{981}{400} > 2,$$

which also contradicts (46). The claimed strict bounds follow. □

We shall use both sides of the annular bound. If $K = \mathbf{Q}(\zeta)$, then the constant term of any positive-degree monic factor of F_ζ in $K[T]$ has absolute value greater than 1 under every embedding of K into $\mathbf{C}$, while each individual root has absolute value less than $5/2$.

9 Two factorization lemmas

Let K be a number field, let $\mathcal{O}_K$ be its ring of integers, and let $\mathfrak{p}$ be a nonzero prime ideal of $\mathcal{O}_K$. Write $v_\mathfrak{p}$ for the additive valuation normalized by $v_\mathfrak{p}(K^\times) = \mathbf{Z}$. A bar denotes reduction modulo $\mathfrak{p}$.

Lemma 9.1. *If a monic polynomial $f \in \mathcal{O}_K[T]$ factors as $f = HJ$ with H, J monic in $K[T]$, then $H, J \in \mathcal{O}_K[T]$.*

Proof. Every root of f is integral over $\mathcal{O}_K$. The coefficients of H and J are elementary symmetric functions in subsets of these roots, hence are integral over $\mathcal{O}_K$. They also lie in K. Since $\mathcal{O}_K$ is integrally closed in K, they belong to $\mathcal{O}_K$. □

Lemma 9.2 (the unsplit-block lemma)**.** *Let $f \in \mathcal{O}_K[T]$ be monic and suppose that, for some integers $a, b \geq 0$ and some $t \in \mathcal{O}_K$ with $\bar{t} \neq 0$,*

$$\bar{f}(T) = T^a (T - \bar{t})^b, \qquad f(t) \in \mathfrak{p} \setminus \mathfrak{p}^2.$$

If $f = HJ$ with $H, J \in \mathcal{O}_K[T]$ monic and $\deg H < b$*, then*

$$\bar{H}(T) = T^{\deg H} \qquad \textit{and} \qquad \deg H \leq a.$$

Proof. Unique factorization over the residue field gives integers r, s such that

$$\bar{H} = T^r (T - \bar{t})^s, \qquad \bar{J} = T^{a-r} (T - \bar{t})^{b-s}.$$

Because H and J are monic, reduction preserves their degrees. If $s > 0$, then $s \leq \deg H < b$, so both s and $b - s$ are positive. Consequently $H(t), J(t) \in \mathfrak{p}$, which would imply $f(t) = H(t)J(t) \in \mathfrak{p}^2$. This is impossible. Thus $s = 0$, so $\bar{H} = T^r$. Degree preservation gives $r = \deg H \leq a$. □

We also use the standard elementary fact that if a sum in a discretely valued field is zero, then its terms cannot have a unique least valuation.

10 Cyclotomic facts at the prime 2

Let ζ have order

$$k = 2^a m, \qquad m \text{ odd},$$

and put $K = \mathbf{Q}(\zeta)$. There is a unique decomposition

$$\zeta = \xi\eta, \tag{47}$$

where ξ has order 2^a and η has order m.

We recall the local cyclotomic facts used below. If $\mathfrak{p} \mid 2$, then reduction is injective on roots of unity of odd order, while every root of unity of 2-power order reduces to 1. The odd-order cyclotomic extension is unramified at 2. If $a \geq 2$, then $\xi - 1$ is a uniformizer and

$$v_{\mathfrak{p}}(2) = 2^{a-1}. \tag{48}$$

Indeed, with $n = 2^{a-1}$, the element $\xi - 1$ is a root of

$$(1 + X)^n + 1,$$

which is Eisenstein at 2. Passing to the compositum with the unramified odd-order part preserves the uniformizer and the ramification index. When $a = 0$, the extension is unramified at 2; when $a = 1$ and m is odd, $\mathbf{Q}(\zeta_{2m}) = \mathbf{Q}(\zeta_m)$, so it is again unramified at 2. These statements are also immediate from the usual decomposition law for primes in cyclotomic fields; see, for example, [Was97, Chapter 2].

11 Orders with a nontrivial odd part

Assume throughout this section that $m > 1$. Fix a prime $\mathfrak{p} \mid 2$ and write

$$e = v_{\mathfrak{p}}(2).$$

Since 4 is invertible modulo m, there is an odd-order root of unity $\theta \in K$ such that $\theta^4 = \eta$. Put

$$t_0 = \theta - 1.$$

The reduction of θ is not 1, so t_0 is a $\mathfrak{p}$-adic unit.

Since $\bar{\xi} = 1$ and $\bar{\eta} \neq 1$, reduction of (42) in characteristic 2 gives

$$\begin{aligned} \overline{F_\zeta(T)} &= T^6 + (\bar{\eta} + 1)T^2 \\ &= T^2\big((T+1)^4 + \bar{\eta}\big) \\ &= T^2(T - \bar{t}_0)^4. \end{aligned} \tag{49}$$

Lemma 11.1. *For every prime* $\mathfrak{p} \mid 2$*,*

$$v_{\mathfrak{p}}\big(F_\zeta(t_0)\big) = 1.$$

Proof. Define the Laurent polynomial

$$B(T) = T^4 + 3T^3 + 5T^2 + 5T - 2 - \frac{4}{T} - \frac{8}{T^2}.$$

The identity

$$g(T) - (T+1)^4 = -2B(T)$$

and $(t_0+1)^4=\eta$ imply
$$F_\eta(t_0)=2t_0^2B(t_0). \tag{50}$$
Modulo $\mathfrak{p}$,
$$\bar{B}(T)=T^4+T^3+T^2+T=T(T+1)^3.$$
This reduction is legitimate because t_0 is a $\mathfrak{p}$-adic unit. Therefore
$$\bar{B}(t_0)=\bar{t}_0\bar{\theta}^3\neq 0,$$
and (50) gives $v_\mathfrak{p}(F_\eta(t_0))=e$.

Moreover,
$$F_\zeta(T)=F_\eta(T)+\eta(\xi-1)T^2. \tag{51}$$
If $a=0$, then $\xi=1$ and $e=1$, so the assertion follows from (50). If $a\geq 2$, the two terms on the right side of (51), evaluated at t_0, have valuations $e\geq 2$ and 1, respectively, so their sum has valuation 1.

It remains to consider $a=1$. Here $\xi=-1$ and $e=1$, whence
$$F_\zeta(t_0)=2t_0^2\big(B(t_0)-\eta\big).$$
After division by 2 and reduction modulo $\mathfrak{p}$,
$$\begin{aligned}\overline{\frac{F_\zeta(t_0)}{2}}&=\bar{t}_0^2\big(\bar{t}_0\bar{\theta}^3+\bar{\theta}^4\big)\\&=\bar{t}_0^2\bar{\theta}^3\neq 0.\end{aligned}$$
Thus the valuation is 1 in every case. □

Proposition 11.2. *If the order of ζ has a nontrivial odd part, then F_ζ is irreducible over K.*

Proof. Suppose that F_ζ is reducible over K. Choose a monic irreducible factor H of minimal degree d, and write $F_\zeta=HJ$ with J monic. Since $\deg F_\zeta=6$, we have $d\leq 3$. By Lemma 9.1, both factors lie in $\mathcal{O}_K[T]$.

At each prime $\mathfrak{p}\mid 2$, equations (49) and Lemma 11.1 allow us to apply Lemma 9.2 with $(a,b)=(2,4)$. Since $d<4$, it follows that $d\leq 2$ and
$$\bar{H}=T^d. \tag{52}$$

First suppose that $d=2$. Put
$$b=H(0),\qquad c=J(0).$$
Then $bc=-16$. At every $\mathfrak{p}\mid 2$, (52) and (49) give
$$\bar{J}=(T-\bar{t}_0)^4,$$
so c is a $\mathfrak{p}$-adic unit. At primes not above 2, the integrality of b,c and $bc=-16$ also show that c is a unit. Hence c is a unit of $\mathcal{O}_K$.

For every complex embedding $\sigma: K\hookrightarrow \mathbf{C}$, the polynomial $\sigma(J)$ is a monic factor of $F_{\sigma(\zeta)}$ of degree 4. By Lemma 8.1, each of its roots has modulus greater than 1. Thus $|\sigma(c)|>1$ for every σ, and therefore
$$|\operatorname{N}_{K/\mathbf{Q}}(c)|=\prod_\sigma|\sigma(c)|>1.$$

This contradicts the fact that the norm of a unit is ± 1.

Now suppose that $d = 1$, say $H(T) = T - \alpha$. From (52), $q = v_{\mathfrak{p}}(\alpha) > 0$ for every $\mathfrak{p} \mid 2$. Moreover, $\zeta - 5$ is a $\mathfrak{p}$-adic unit, because $\bar{\zeta} = \bar{\eta} \neq 1$. The valuations of the seven terms in $F_\zeta(\alpha) = 0$, ordered from the constant term to the leading term, are

$$4e, \quad 3e + q, \quad 2q, \quad e + 3q, \quad 2e + 4q, \quad e + 5q, \quad 6q.$$

If $q < 2e$, then $2q$ is the unique least value. If $q > 2e$, then $4e$ is the unique least value. Since a zero sum cannot have a unique term of least valuation, necessarily

$$v_{\mathfrak{p}}(\alpha) = 2e = v_{\mathfrak{p}}(4) \qquad (\mathfrak{p} \mid 2). \tag{53}$$

At every prime away from 2, the equality $(-\alpha)J(0) = -16$ and integrality give valuation zero for α. Thus $(\alpha) = (4)$, and $u = \alpha/4$ is a unit of $\mathcal{O}_K$.

For every complex embedding σ, Lemma 8.1 gives

$$|\sigma(u)| = \frac{|\sigma(\alpha)|}{4} < \frac{5}{8}.$$

It follows that $|\,\mathrm{N}_{K/\mathbf{Q}}(u)| < 1$, contradicting that u is a unit. Both possible degrees have been excluded, so F_ζ is irreducible. $\square$

12 Pure powers of 2

Assume now that ζ has order 2^a with $a \geq 1$. There is a unique prime $\mathfrak{p}$ of K above 2, its residue degree is 1, and

$$e = v_{\mathfrak{p}}(2) = \varphi(2^a).$$

For $a = 1$, $K = \mathbf{Q}$ and $\zeta = -1$. For $a \geq 2$, $\zeta - 1$ is a uniformizer. In all cases,

$$v_{\mathfrak{p}}(\zeta - 5) = 1. \tag{54}$$

Indeed, for $a = 1$ one has $\zeta - 5 = -6$; for $a \geq 2$ one has $\zeta - 5 = (\zeta - 1) - 4$, whose two summands have valuations 1 and $2e > 1$.

Reduction modulo $\mathfrak{p}$ gives

$$\overline{F_\zeta(T)} = T^6. \tag{55}$$

Proposition 12.1. *If ζ has 2-power order and $\zeta \neq 1$, then F_ζ is irreducible over K.*

Proof. Suppose that F_ζ is reducible. Choose a monic irreducible factor H of minimal degree $d \leq 3$, and write $F_\zeta = HJ$ with J monic. Lemma 9.1 gives $H, J \in \mathcal{O}_K[T]$, and (55) gives

$$\bar{H} = T^d, \qquad \bar{J} = T^{6-d}. \tag{56}$$

We first exclude $d = 1$. Write $H = T - \alpha$. Then $q = v_{\mathfrak{p}}(\alpha)$ is a positive integer. The valuations of the terms of $F_\zeta(\alpha) = 0$, from constant to leading, are

$$4e, \quad 3e + q, \quad 1 + 2q, \quad e + 3q, \quad 2e + 4q, \quad e + 5q, \quad 6q.$$

If $q \leq 2e - 1$, then $1 + 2q$ is the unique least value. If $q \geq 2e$, then $4e$ is the unique least value. Both cases contradict the nonarchimedean triangle inequality. Hence $d \neq 1$.

Next suppose that $d = 3$. Write

$$H = T^3 + h_2 T^2 + h_1 T + h_0, \qquad J = T^3 + j_2 T^2 + j_1 T + j_0.$$

By (56), every displayed coefficient other than the leading coefficients belongs to $\mathfrak{p}$. The coefficient of T^2 in HJ is

$$h_2 j_0 + h_1 j_1 + h_0 j_2 \in \mathfrak{p}^2.$$

But the coefficient of T^2 in F_ζ is $\zeta - 5$, which belongs to $\mathfrak{p} \setminus \mathfrak{p}^2$ by (54). This contradiction excludes $d = 3$.

Consequently $d = 2$. Write

$$H = T^2 + h_1 T + h_0, \qquad J = T^4 + j_3 T^3 + j_2 T^2 + j_1 T + j_0.$$

All the h_i and j_i lie in $\mathfrak{p}$. Comparing the coefficient of T^2 gives

$$\zeta - 5 = j_0 + h_1 j_1 + h_0 j_2 \equiv j_0 \pmod{\mathfrak{p}^2}. \tag{57}$$

Therefore $j_0 \in \mathfrak{p} \setminus \mathfrak{p}^2$.

Put $b = H(0)$ and $c = J(0) = j_0$. At every odd prime, the integrality of b, c and $bc = -16$ implies that c is a unit. Since $\mathfrak{p}$ is the unique prime above 2, (57) shows that

$$(c) = \mathfrak{p}.$$

The residue degree of $\mathfrak{p}$ is 1, so

$$|\operatorname{N}_{K/\mathbf{Q}}(c)| = \operatorname{N}(\mathfrak{p}) = 2. \tag{58}$$

For every complex embedding σ, the two roots of $\sigma(H)$ have modulus less than $5/2$ by Lemma 8.1. Since the absolute value of the product of all six roots of $F_{\sigma(\zeta)}$ is 16,

$$|\sigma(c)| = \frac{16}{|\sigma(b)|} > \frac{16}{(5/2)^2} = \frac{64}{25}.$$

Taking the product over all embeddings gives

$$|\operatorname{N}_{K/\mathbf{Q}}(c)| > \left(\frac{64}{25}\right)^{[K:\mathbf{Q}]} > 2,$$

contradicting (58). Thus F_ζ is irreducible. □

Theorem 12.2 (Cyclotomic-fiber irreducibility)**.** *For every root of unity $\zeta \neq 1$, the polynomial F_ζ is irreducible over $\mathbf{Q}(\zeta)$.*

Proof. Proposition 11.2 applies when the order of ζ has a nontrivial odd part, and Proposition 12.1 applies to the remaining nontrivial roots of unity. □

Theorem 1.2 now follows from Proposition 7.1.

13 The period-4 multiplier curve and the delta factors

We use the standard notation of [Mor96, MV95, KMST25]. Let $\delta_m(x, c)$ denote the multiplier polynomial for cycles of formal period m, put $C = 4c$, and set

$$\widetilde{\delta}_m(x, C) = \delta_m(x, C/4).$$

The multiplier curve X_m is the affine curve $\widetilde{\delta}_m(x, C) = 0$. A direct computation from the definition in [KMST25, Section 2] gives, in period 4,

$$\begin{aligned} \widetilde{\delta}_4(x, C) = C^6 + 12C^5 + (x + 48)C^4 + (4x + 192)C^3 \\ + (512 - 16x - x^2)C^2 + (16 - x)^3. \end{aligned} \tag{59}$$

Proposition 13.1 (Normalization of the period-4 multiplier curve)**.** *The morphism*

$$\nu : \mathbf{G}_m \longrightarrow X_4, \qquad T \longmapsto \big(g(T), h(T)\big), \qquad h(T) = -T^2 - 3 - \frac{4}{T},$$

is the normalization of X_4. In particular, it is finite, birational, and surjective.

Proof. Exact substitution in (59) gives

$$\widetilde{\delta}_4\big(g(T), h(T)\big) = 0. \tag{60}$$

This Laurent-polynomial identity is also checked coefficient by coefficient by the certificate in Appendix A.

Put

$$A = \mathbf{Q}[x, C]/\big(\widetilde{\delta}_4(x, C)\big), \qquad B = \mathbf{Q}[T, T^{-1}].$$

As a rational map $\mathbf{P}^1 \to \mathbf{P}^1$, the function

$$h(T) = -\frac{T^3 + 3T + 4}{T}$$

has degree 3. Thus

$$[\overline{\mathbf{Q}}(T) : \overline{\mathbf{Q}}(h)] = 3.$$

We claim that $g \notin \overline{\mathbf{Q}}(h)$. Otherwise $g = R(h)$ for some $R \in \overline{\mathbf{Q}}(X)$. Since g has no poles outside 0 and ∞, while every finite value of h has all its preimages in $\mathbf{G}_m$, the function R has no finite pole and is a polynomial. The pole orders at infinity force $\deg R = 2$. There the leading terms $h(T)^2 \sim T^4$ and $g(T) \sim -T^4$ force the leading coefficient of R to be -1; at zero, the leading terms $h(T)^2 \sim 16T^{-2}$ and $g(T) \sim 16T^{-2}$ force it to be 1. This contradiction proves the claim.

The degree-three extension above has no nontrivial intermediate field, so

$$\overline{\mathbf{Q}}(g, h) = \overline{\mathbf{Q}}(T).$$

Since (59) has degree three in x and vanishes at (g, h), it is the minimal polynomial of g over $\overline{\mathbf{Q}}(h)$ up to a nonzero constant. Its leading coefficient as a polynomial in x is -1, so Gauss's lemma makes it absolutely irreducible. Moreover, $g \notin \overline{\mathbf{Q}}(h)$ implies $g \notin \mathbf{Q}(h)$; since $[\mathbf{Q}(T) : \mathbf{Q}(h)] = 3$, the same prime-degree argument gives $\mathbf{Q}(g, h) = \mathbf{Q}(T)$. Consequently A is a domain, and the homomorphism $A \to B$ defined by (60) is injective and birational. In B one has

$$T^3 + (C + 3)T + 4 = 0, \qquad T^{-1} = -\frac{T^2 + C + 3}{4}.$$

Thus $B = A[T]$ is finite over A. Since B is normal, it is the integral closure of A in their common function field. Hence $\operatorname{Spec} B \to \operatorname{Spec} A$ is the normalization; being finite and dominant, it is surjective. □

For $k \geq 2$, define

$$\widetilde{\Delta}_{mk,m}(C) = \Delta_{mk,m}(C/4) = \operatorname{Res}_x\big(\Phi_k(x), \widetilde{\delta}_m(x, C)\big).$$

Koizumi, Murakami, Sano, and Takehira prove that these delta factors are separable and that projection to the parameter coordinate gives the relevant Galois-equivariant root correspondence [KMST25, Proposition 2.3 and Section 2]. More precisely, if $Z(q)$ denotes the set of roots of q in $\overline{\mathbf{Q}}$, without multiplicity, then

$$\big\{(\lambda, C) \in X_m(\overline{\mathbf{Q}}) : \lambda \text{ is a primitive } k\text{-th root of unity}\big\} \xrightarrow{\sim} Z(\widetilde{\Delta}_{mk,m}). \tag{61}$$

Their degree formula, originating in [MV95, Corollary 3.3], is

$$\deg_C \widetilde{\Delta}_{mk,m} = \nu(m)\varphi(k), \qquad \nu(m) = \sum_{d|m} 2^{d-1}\mu(m/d). \tag{62}$$

Here μ is the Möbius function. In particular, $\nu(4) = 0 - 2 + 8 = 6$.

Lemma 13.2 (Root correspondence in period 4)**.** *For every $k \geq 2$, the map*

$$\beta : Z(\Gamma_k) \longrightarrow Z(\widetilde{\Delta}_{4k,4}), \qquad \beta(\alpha) = h(\alpha),$$

is a bijection equivariant for $\mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$*.*

Proof. The polynomial Γ_k is irreducible of degree $6\varphi(k)$. Its constant term is $(-16)^{\varphi(k)}$, so its distinct roots lie in $\mathbf{G}_m$. Proposition 13.1 maps them onto

$$\big\{(\lambda, C) \in X_4(\overline{\mathbf{Q}}) : \lambda \text{ is a primitive } k\text{-th root of unity}\big\}.$$

Indeed, surjectivity of the normalization supplies a preimage $T \in \overline{\mathbf{Q}}^\times$ of every point in this set, and then $\Phi_k(g(T)) = 0$; conversely every root of Γ_k supplies such a point. By (61), separability, and (62), both the source and target have $6\varphi(k)$ elements. The resulting surjection is therefore a bijection. It is Galois-equivariant because g and h are defined over $\mathbf{Q}$. □

The absolute Galois group acts transitively on $Z(\Gamma_k)$ by Theorem 1.2. Lemma 13.2 and the separability of $\widetilde{\Delta}_{4k,4}$ therefore prove that $\widetilde{\Delta}_{4k,4}$ is irreducible for $k \geq 2$.

For $k = 1$, the table in [MV95, Section 3, Table 1] gives

$$\widetilde{\delta}_4(1, C) = (C^2 - 2C + 5)(C + 5)(C^3 + 9C^2 + 27C + 135).$$

Since $\widetilde{\delta}_1(x, C) = x^2 - 2x + C$ and $\widetilde{\delta}_2(x, C) = C + 4 - x$, the two lower-period factors are

$$\widetilde{\Delta}_{4,1}(C) = \mathrm{Res}_x\big(x^2 + 1, \widetilde{\delta}_1(x, C)\big) = C^2 - 2C + 5, \qquad \widetilde{\Delta}_{4,2}(C) = \widetilde{\delta}_2(-1, C) = C + 5.$$

Consequently the defining product decomposition $\widetilde{\delta}_4(1, C) = \widetilde{\Delta}_{4,1}(C)\widetilde{\Delta}_{4,2}(C)\widetilde{\Delta}_{4,4}(C)$ gives

$$\widetilde{\Delta}_{4,4}(C) = C^3 + 9C^2 + 27C + 135 = (C + 3)^3 + 108.$$

A rational root would be an integer and would make an integer cube equal to -108, which is impossible because $v_2(108) = 2$ is not divisible by 3. Thus the cubic is irreducible. The invertible linear change $C = 4c$ preserves irreducibility over $\mathbf{Q}$, completing the proof of Corollary 1.5.

Remark 13.3 (Scope of the rational parametrization)**.** *For the quadratic family, the normalizations of the multiplier curves X_m are rational exactly for $m \leq 4$. For $m = 1, 2$ this follows from the displayed elementary equations; period 3 is parametrized in [KMST25, Lemma 2.6], and period 4 is Proposition 13.1. For $m \geq 5$, Morton's genus formula and its quadratic-family corollary give genus at least two [Mor96, Theorem 13(b),(c) and the corollary following Theorem 14]. Morton's calculation is made under his hypothesis*

(H). Its clause asserting simple roots at multiplier 1 is justified there using the Douady–Hubbard description of hyperbolic components; see the discussion preceding hypothesis (H) in [Mor96]. The local simplification used above also exploits fourth powers in residue characteristic two. Thus period 4 is the endpoint reached by the present implementation of the rational-parametrization method; this is not an obstruction to different methods at higher periods.

Part III

Sextic monodromy and arithmetic base change

14 Newton polygons and a denominator lemma

We first record precisely the local consequence of Newton polygons that will be used below. If L is a complete discretely valued field, its valuation is normalized by $v(L^\times) = \mathbf{Z}$, and the same letter v denotes a fixed extension to $\overline{L}$. We use the standard Newton polygon theorem over complete discretely valued fields [Neu99, Chapter II, §6, Propositions (6.3) and (6.4)].

Lemma 14.1 (Newton-polygon denominator lemma)**.** *Let $P \in L[T]$ be monic with $P(0) \neq 0$. Suppose that the Newton polygon of P has an edge of horizontal length ℓ and slope*

$$-\frac{h}{d}, \qquad h \in \mathbf{Z}, \quad d \geq 1, \quad \gcd(h, d) = 1.$$

Then the slope factor belonging to that edge has degree ℓ, and every irreducible factor of the slope factor has degree divisible by d. In particular, if $\ell = d$, the slope factor is irreducible of degree d over L.

Proof. By the Newton polygon theorem over the complete, hence henselian, field L, the polynomial P has a factor $P_{h/d} \in L[T]$ whose roots, counted with multiplicity, are exactly the ℓ roots β of P for which

$$v(\beta) = \frac{h}{d}.$$

Thus $\deg P_{h/d} = \ell$.

Let $G \in L[T]$ be an irreducible factor of $P_{h/d}$ and let β be a root of G. Put $M = L(\beta)$ and let $e(M/L)$ be the ramification index. With the normalization $v(L^\times) = \mathbf{Z}$, the value group of M is

$$v(M^\times) = \frac{1}{e(M/L)}\mathbf{Z}.$$

Since $v(\beta) = h/d$ is in lowest terms, it follows that $d \mid e(M/L)$. The ramification index divides $[M : L] = \deg G$, so $d \mid \deg G$. If $\ell = d$, the degrees of the irreducible factors of $P_{h/d}$ are positive multiples of d whose sum is d; hence there is exactly one such factor. □

15 A local quartic factor of the period-4 fiber

Let $\zeta \neq 1$ be a root of unity, put $K = \mathbf{Q}(\zeta)$, and write

$$\operatorname{ord}(\zeta) = 2^a m, \qquad m \text{ odd}, \qquad \zeta = \xi\eta,$$

where ξ has 2-power order and η has odd order. Recall that

$$F_\zeta(T) = T^6 + 2T^5 + 4T^4 + 6T^3 + (\zeta - 5)T^2 - 8T - 16.$$

Fix a prime $\mathfrak{p}$ of K above 2, write $K_\mathfrak{p}$ for the completion, normalize $v = v_\mathfrak{p}$ by $v(K_\mathfrak{p}^\times) = \mathbf{Z}$, and put $e = v(2)$.

When $m > 1$, choose the odd-order root of unity $\theta \in K$ used in Section 11, so that $\theta^4 = \eta$, and put $t_0 = \theta - 1$. The base irreducibility argument already proved that t_0 is a $\mathfrak{p}$-adic unit and that

$$\overline{F_\zeta(T)} = T^2(T - \overline{t_0})^4, \qquad v\big(F_\zeta(t_0)\big) = 1;$$

see (49) and Lemma 11.1.

Proposition 15.1 (Local quartic factor)**.** *For every root of unity $\zeta \neq 1$, the polynomial F_ζ has an irreducible factor of degree 4 over $K_\mathfrak{p}$.*

Proof. Suppose first that $m > 1$. Write

$$F_\zeta(t_0 + Y) = \sum_{j=0}^{6} b_j Y^j.$$

Lemma 11.1 gives $v(b_0) = 1$. Reducing the shifted polynomial and using (49), we obtain, in residue characteristic 2,

$$\begin{aligned}\overline{F_\zeta(t_0 + Y)} &= (Y + \overline{t_0})^2 Y^4 \\ &= Y^6 + \overline{t_0}^2 Y^4.\end{aligned}$$

Consequently

$$v(b_j) \geq 1 \quad (1 \leq j \leq 3), \qquad v(b_4) = 0, \qquad v(b_5) \geq 1, \qquad v(b_6) = 0.$$

Every point $(j, v(b_j))$ lies on or above the line $y = 1 - j/4$, and equality holds at $j = 0$ and $j = 4$. Hence the Newton polygon contains the edge

$$(0,1) \longrightarrow (4,0),$$

of length 4 and slope $-1/4$. Lemma 14.1 shows that the corresponding degree-four slope factor of $F_\zeta(t_0 + Y)$ is irreducible over $K_\mathfrak{p}$. Translating $Y = T - t_0$ preserves degrees and irreducibility, and therefore gives the required factor of F_ζ.

Now suppose that $m = 1$. Since $\zeta \neq 1$, its order is 2^a with $a \geq 1$. The recalled local cyclotomic facts give

$$e = v(2) = 2^{a-1}.$$

For $a = 1$ one has $\zeta = -1$ and $v(\zeta - 5) = v(-6) = 1$. For $a \geq 2$, the element $\zeta - 1$ is a uniformizer, and

$$\zeta - 5 = (\zeta - 1) - 4$$

is a sum of terms of respective valuations 1 and $2e > 1$; hence again $v(\zeta - 5) = 1$. It follows directly from the seven coefficients of F_ζ, listed from the constant coefficient to the leading coefficient, that their valuations are

$$(4e,\ 3e,\ 1,\ e,\ 2e,\ e,\ 0). \tag{63}$$

The lower convex hull of these seven points is

$$(0, 4e) \longrightarrow (2, 1) \longrightarrow (6, 0). \tag{64}$$

Indeed, the point $(1, 3e)$ lies strictly above the first segment because

$$3e > \frac{4e + 1}{2},$$

while the extension of that segment has negative ordinate at every integer abscissa $j \geq 3$, since at $j = 3$ its ordinate is $-2e + 3/2 < 0$. All the remaining coefficient valuations are nonnegative. On the other hand, the points with abscissas 0 and 1 lie strictly above the line through $(2, 1)$ and $(6, 0)$, whose ordinates there are $3/2$ and $5/4$, respectively, because $4e > 3/2$ and $3e > 5/4$. Finally, the points with abscissas $3, 4, 5$ lie strictly above the second segment, whose ordinates there are $3/4, 1/2, 1/4$, respectively. The two displayed slopes are

$$-\frac{4e - 1}{2} \qquad \text{and} \qquad -\frac{1}{4},$$

and are strictly increasing. These observations also show directly that no omitted point lies below either segment. The second edge in (64) has length 4 and reduced slope $-1/4$. Lemma 14.1 therefore makes its slope factor an irreducible quartic over $K_{\mathfrak{p}}$. $\square$

16 A sextic criterion for abelian disjointness

Let k be a number field and let

$$f(T) = \prod_{i=1}^{6}(T - r_i) \in k[T]$$

be an irreducible separable sextic. For each unordered partition $\{A, A^c\}$ of $\{1, \ldots, 6\}$ into two triples, put

$$s_A = \prod_{i \in A} r_i + \prod_{j \in A^c} r_j.$$

There are $\binom{6}{3}/2 = 10$ such partitions. Define the $3 + 3$ resolvent by

$$\mathcal{R}_f(S) = \prod_{\{A, A^c\}} (S - s_A). \tag{65}$$

The absolute Galois group of k permutes the ten unordered partitions; thus $\mathcal{R}_f(S) \in k[S]$.

Write k^{ab} for the maximal abelian extension of k in a fixed algebraic closure.

Theorem 16.1 (Abelian-disjointness criterion for sextics)**.** *Let α be a root of f and put $E = k(\alpha)$. Suppose that*

(i) *for some finite place v of k, the polynomial f has an irreducible quartic factor over k_v;*

(ii) *the resolvent $\mathcal{R}_f$ has no root in k.*

Then

$$k \subseteq D \subseteq E, \quad D/k \text{ Galois} \quad \Longrightarrow \quad D = k.$$

In particular,

$$E \cap k^{\mathrm{ab}} = k.$$

Consequently f remains irreducible over k^{ab}.

Proof. Let D be an intermediate field with D/k Galois. Since f is irreducible of degree six, $[E:k]=6$, and the tower $k \subseteq D \subseteq E$ gives

$$[D:k] \mid 6.$$

It remains to exclude the three possibilities $[D:k]=2,3,6$.

Suppose first that $[D:k]=2$, and let τ be the nontrivial element of $\mathrm{Gal}(D/k)$. The minimal polynomial H of α over D is monic of degree $[E:D]=3$ and divides f in $D[T]$. The polynomials H and $\tau(H)$ are distinct: otherwise H would have coefficients fixed by τ, hence would lie in $k[T]$, contradicting the irreducibility of the degree-six polynomial f over k. Both are irreducible over D, so they are coprime. Their product is monic of degree six, divides f, and is fixed by τ; therefore

$$f = H\,\tau(H).$$

The roots of H form a triple $\{r_i : i \in A\}$ and those of $\tau(H)$ form the complementary triple. Thus

$$H(0) = -\prod_{i\in A} r_i, \qquad \tau(H)(0) = -\prod_{j\in A^c} r_j,$$

and consequently

$$-\operatorname{Tr}_{D/k}\big(H(0)\big) = \prod_{i\in A} r_i + \prod_{j\in A^c} r_j = s_A.$$

This is a root of $\mathcal{R}_f$ lying in k, contrary to hypothesis (ii). Hence $[D:k]\neq 2$.

Suppose next that $[D:k]=3$. Let $q \in k_v[T]$ be an irreducible quartic factor of f, let β be a root of q, and choose a k-embedding

$$\sigma : E \hookrightarrow \overline{k_v} \qquad \text{with} \qquad \sigma(\alpha)=\beta.$$

Such an embedding exists because β is a root of the minimal polynomial f of α. Put $L = k_v(\beta)$, so $[L:k_v]=4$. The compositum $k_v\sigma(D) \subset \overline{k_v}$ is canonically isomorphic to the completion D_w at the place w induced by σ, and it is contained in L because $D \subseteq E = k(\alpha)$.

Since D/k is Galois of prime degree three, the local degree $[D_w : k_v]$ is either 1 or 3. If it is 3, then the tower

$$k_v \subseteq k_v\sigma(D) \subseteq L$$

would force 3 to divide $[L:k_v]=4$, which is impossible. If it is 1, then v splits completely in D and $\sigma(D) \subseteq k_v$. The minimal polynomial of α over D has degree $[E:D]=2$; applying σ to its coefficients gives a quadratic in $k_v[T]$ having β as a root. This implies $[k_v(\beta):k_v] \le 2$, again contradicting $[L:k_v]=4$. Thus $[D:k]\neq 3$.

Finally, suppose that $[D:k]=6$. The inclusions $D\subseteq E$ and the equality of degrees give $D=E$. In particular, E/k is Galois. With q, β, and σ as in the preceding paragraph, we have

$$k_v(\beta) = k_v\sigma(E),$$

because $E=k(\alpha)$. The field on the right is a completion of the Galois extension E/k, so its degree over k_v is the order of a decomposition group and therefore divides $[E:k]=6$. Its degree is also 4, by the irreducibility of q. This contradiction excludes $[D:k]=6$.

We have proved that E contains no nontrivial subextension Galois over k. Now $E\cap k^{\mathrm{ab}}$ is finite and Galois over k, so it equals k. It remains to justify the assertion over the infinite extension k^{ab}. If f were reducible over k^{ab}, a nontrivial factorization would involve only finitely many coefficients. They would all lie in some finite intermediate extension

$$k \subseteq M \subseteq k^{\mathrm{ab}}.$$

The extension M/k is finite abelian Galois, and

$$E \cap M \subseteq E \cap k^{\mathrm{ab}} = k.$$

Thus $E \cap M = k$. Since M/k is Galois, the fields E and M are linearly disjoint over k. More explicitly, restriction identifies $\mathrm{Gal}(EM/E)$ with $\mathrm{Gal}(M/E \cap M) = \mathrm{Gal}(M/k)$; hence $[EM : E] = [M : k]$, and the tower formula gives

$$[M(\alpha) : M] = [E : k] = 6.$$

The minimal polynomial of α over M therefore still has degree six and equals f, contradicting the assumed factorization over M. Hence f is irreducible over k^{ab}. $\square$

17 The 3+3 resolvent

Put

$$P_X(T) = T^6 + 2T^5 + 4T^4 + 6T^3 + (X-5)T^2 - 8T - 16$$

so that $P_\zeta(T) = F_\zeta(T)$. Let $r_1, \ldots, r_6$ be the roots of P_X in a splitting field over $\mathbf{Q}(X)$. If $A \subset \{1, \ldots, 6\}$ has cardinality three, write $y_A = \prod_{i \in A} r_i$. For an unordered complementary pair $\{A, A^c\}$, put $s_A = y_A + y_{A^c}$. There are ten such pairs, and we define

$$\mathcal{R}_X(S) = \prod_{\{A, A^c\}} (S - s_A).$$

The coefficients are symmetric polynomials in the r_i with integer coefficients. The fundamental theorem on symmetric polynomials therefore gives $\mathcal{R}_X \in \mathbf{Z}[X, S]$.

Let C_X be the companion matrix of P_X, in a convention for which its characteristic polynomial is P_X, and put

$$Q_X(Y) = \det\Big(YI - \bigwedge^3 C_X\Big).$$

The eigenvalues of $\bigwedge^3 C_X$ are the twenty numbers y_A. Moreover,

$$y_A y_{A^c} = r_1 \cdots r_6 = -16, \qquad y_{A^c} = -\frac{16}{y_A}.$$

Consequently, for each complementary pair,

$$(Y - y_A)(Y - y_{A^c}) = Y^2 - s_A Y - 16,$$

and hence

$$Q_X(Y) = Y^{10} \mathcal{R}_X\left(Y - \frac{16}{Y}\right). \tag{66}$$

Exact expansion gives

$$\begin{aligned}\mathcal{R}_X(S) = S^{10} &+ 6S^9 + 4(X+35)S^8 + 2(X^2 - 6X + 389)S^7 \\ &+ (X^3 - 15X^2 + 667X + 5683)S^6 + 96(X^2 - 2X + 337)S^5 \\ &+ 64(X^3 - 14X^2 + 477X + 800)S^4 - 1536(X^2 - 21X - 368)S^3 \\ &+ 8192(X^2 + 35X - 48)S^2 - 65536(X - 16)(X + 4)S \\ &- 65536(X - 16)^2.\end{aligned} \tag{67}$$

Identity (67) is certified independently by the exact-arithmetic exterior-power computation described in Appendix A; no numerical specialization is used.

18 Absence of K-rational roots of the resolvent

Let $\zeta \neq 1$ be a root of unity, put $K = \mathbf{Q}(\zeta)$, and set $\mathcal{R}_\zeta = \mathcal{R}_X|_{X=\zeta}$. Fix a prime $\mathfrak{p} \mid 2$, normalize $v = v_\mathfrak{p}$ by $v(K_\mathfrak{p}^\times) = \mathbf{Z}$, and put $e = v(2)$. Write $\bar{x}$ for the reduction of an integral element x modulo $\mathfrak{p}$.

18.1 Orders with a nontrivial odd part

Write $\operatorname{ord}(\zeta) = 2^a m$ with $m > 1$ odd, and decompose $\zeta = \xi\eta$, where ξ has 2-power order and η has odd order. Since $\bar{\xi} = 1$ and reduction is injective on odd-order roots of unity, $\bar{\zeta} = \bar{\eta} \neq 1$.

Write $\mathcal{R}_\zeta(S) = \sum_{j=0}^{10} c_j S^j$. Reducing the unit part of each coefficient in (67) gives the following exact valuations:

$$\big(v(c_0), \dots, v(c_{10})\big) = e(16, 16, 13, 9, 6, 5, 0, 1, 2, 1, 0).$$

For completeness, the reductions that prove exactness are

$$\begin{aligned}\Big(c_j/2^{v(c_j)/e} \bmod \mathfrak{p}\Big)_{j=0}^{10} = \big(&\bar{\zeta}^2, \bar{\zeta}^2, \bar{\zeta}(\bar{\zeta}+1), \bar{\zeta}(\bar{\zeta}+1),\\ &\bar{\zeta}(\bar{\zeta}+1)^2, (\bar{\zeta}+1)^2, (\bar{\zeta}+1)^3, (\bar{\zeta}+1)^2,\\ &\bar{\zeta}+1, 1, 1\big).\end{aligned}$$

Odd rational-integer factors reduce to 1 and have therefore been suppressed; every displayed residue is nonzero.

The lower convex hull is

$$(0, 16e) \longrightarrow (6, 0) \longrightarrow (10, 0). \tag{68}$$

Indeed, after division by e, the differences between the valuations at abscissas $1, \dots, 5$ and the line from $(0, 16e)$ to $(6, 0)$ are

$$\frac{8}{3}, \quad \frac{7}{3}, \quad 1, \quad \frac{2}{3}, \quad \frac{7}{3},$$

all positive. From abscissa 6 onward the lower boundary is horizontal at height 0. The first six roots thus have valuation $8e/3$. The ramification index e is a power of two, so $8e/3 \notin \mathbf{Z}$; none of these six roots belongs to $K_\mathfrak{p}$.

It remains to exclude the four unit roots. Since $\gcd(4, m) = 1$, the map $x \mapsto x^4$ is an automorphism of the cyclic group $\langle\eta\rangle$. Let $\theta \in \langle\eta\rangle$ be the unique element satisfying $\theta^4 = \eta$, and put

$$s_0 = (\theta + 1)^3.$$

Since $\theta \neq 1$ and reduction is injective on odd-order roots of unity, $\bar{\theta} \neq 1$; hence s_0 is a $\mathfrak{p}$-adic unit. Only the coefficients of S^{10} and S^6 survive modulo $\mathfrak{p}$, and

$$\overline{c_6} = (\bar{\zeta} + 1)^3 = (\bar{\theta} + 1)^{12} = \bar{s}_0^4.$$

Thus

$$\overline{\mathcal{R}_\zeta(S)} = S^6 (S - \bar{s}_0)^4. \tag{69}$$

We shall use the following three coefficientwise congruences in $\mathbf{Z}[u]$:

$$\mathcal{R}_{u^4}\big((u+1)^3\big) \equiv 2u^3(u+1)^{27} \pmod 4, \tag{70}$$
$$\frac{\partial \mathcal{R}_X}{\partial X}\big((u+1)^3\big)\Big|_{X=u^4} \equiv (u+1)^{26} \pmod 2, \tag{71}$$
$$\mathcal{R}_{-u^4}\big((u+1)^3\big) \equiv 2u^3(u+1)^{26} \pmod 4. \tag{72}$$

They are verified by the exact-arithmetic certificate described in Appendix A, which checks every coefficient of the three differences.

We now prove
$$v\big(\mathcal{R}_\zeta(s_0)\big) = 1. \tag{73}$$
If $a=0$, then $\zeta=\eta=\theta^4$ and $e=1$; congruence (70) writes $\mathcal{R}_\zeta(s_0)$ as twice a unit plus an element of $4\mathcal{O}_{K_\mathfrak{p}}$, and proves (73). If $a=1$, then $\zeta=-\eta=-\theta^4$ and $e=1$, so (72) gives the same conclusion in the same way.

Suppose $a\geq 2$. Set $\delta=\eta(\xi-1)$, so that $\zeta=\eta+\delta$ and $v(\delta)=1$. Taylor expansion in the polynomial variable X gives

$$\mathcal{R}_{\eta+\delta}(s_0) = \mathcal{R}_\eta(s_0) + \delta\,\partial_X \mathcal{R}_X(s_0)|_{X=\eta} + \delta^2 A$$

for some $A\in\mathcal{O}_{K_\mathfrak{p}}$; its integrality follows because $\mathcal{R}_X(s_0)$ has coefficients in $\mathcal{O}_{K_\mathfrak{p}}$. Congruence (70) gives $v(\mathcal{R}_\eta(s_0)) = e \geq 2$, while (71) says that the displayed derivative is a unit. The linear term therefore has valuation one and every other term has valuation at least two, proving (73) without cancellation.

After the translation $S=s_0+Y$, all coefficients remain integral, and equation (69) shows that the coefficients of Y, Y^2, Y^3 have positive valuation and that the coefficient of Y^4 is a unit. Together with (73), this puts the edge

$$(0,1) \longrightarrow (4,0)$$

in the Newton polygon. Its slope is $-1/4$ and its horizontal length is four. Lemma 14.1 therefore makes the corresponding degree-four factor irreducible. In particular, none of these four roots lies in $K_\mathfrak{p}$. Together with (68), this proves that $\mathcal{R}_\zeta$ has no root in K whenever $m>1$.

18.2 Pure powers of two

First let $\zeta=-1$. Substitution in (67) gives $c_j = 2^{v(c_j)}u_j$, with the following odd unit parts u_j:

j	0	1	2	3	4	5	6	7	8	9	10
u_j	-289	51	-41	519	77	255	625	99	17	3	1,

and therefore
$$\big(v(c_0),\ldots,v(c_{10})\big) = (16,16,14,10,8,7,3,3,3,1,0).$$
The differences above the line from $(0,16)$ to $(6,3)$ at abscissas $1,\ldots,5$ are

$$\frac{13}{6},\quad \frac{7}{3},\quad \frac{1}{2},\quad \frac{2}{3},\quad \frac{11}{6}.$$

The points at abscissas $7,8,9$ also lie strictly above the line from $(6,3)$ to $(10,0)$. Hence the lower convex hull is
$$(0,16) \longrightarrow (6,3) \longrightarrow (10,0),$$

with slopes $-13/6$ and $-3/4$. The six roots belonging to the first edge have valuation $13/6$, and the four belonging to the second have valuation $3/4$. Neither valuation is integral. In particular, $\mathcal{R}_{-1}$ has no root in $\mathbf{Q}_2$.

Now suppose that ζ has order 2^a with $a \geq 2$. Put $\pi = \zeta - 1$ and $e = v(2) = 2^{a-1}$, so $v(\pi) = 1$. Substituting $X = 1 + \pi$ into the parenthesized coefficient polynomials in (67) gives

$$\begin{aligned} X^2 + 35X - 48 &= \pi^2 + 37\pi - 12, \\ X^2 - 21X - 368 &= \pi^2 - 19\pi - 388, \\ X^3 - 14X^2 + 477X + 800 &= \pi^3 - 11\pi^2 + 452\pi + 1264, \\ X^2 - 2X + 337 &= \pi^2 + 336, \\ X^3 - 15X^2 + 667X + 5683 &= \pi^3 - 12\pi^2 + 640\pi + 6336, \\ X^2 - 6X + 389 &= \pi^2 - 4\pi + 384, \\ X + 35 &= \pi + 36. \end{aligned}$$

In addition, $X - 16 = \pi - 15$ and $X + 4 = \pi + 5$ are units. Since $e \geq 2$, each displayed expression has a unique term of least valuation. It follows that

$$\begin{aligned} \big(v(c_0), \ldots, v(c_{10})\big) = (&16e, 16e, 13e+1, 9e+1, 6e+2, 5e+2, \\ &3, e+2, 2e+1, e, 0). \end{aligned} \tag{74}$$

For the first segment, the differences at abscissas $1, \ldots, 5$ are

$$\frac{16e-3}{6}, \quad \frac{7e}{3}, \quad \frac{2e-1}{2}, \quad \frac{2e}{3}, \quad \frac{14e-3}{6},$$

all positive. At abscissas $7, 8, 9$, the valuations $e+2, 2e+1, e$ are strictly greater than the respective heights $9/4, 3/2, 3/4$ of the second segment. Thus the lower convex hull is

$$(0, 16e) \longrightarrow (6, 3) \longrightarrow (10, 0),$$

with slopes

$$-\frac{16e-3}{6}, \qquad -\frac{3}{4}.$$

The integer $16e - 3$ is odd, and modulo three it is congruent to e. Since e is a power of two, $3 \nmid e$; hence

$$\gcd(16e - 3, 6) = 1.$$

The two slopes have reduced denominators six and four, respectively, and the corresponding horizontal lengths are six and four. Lemma 14.1 therefore shows that the two slope factors are irreducible of degrees six and four. In particular, $\mathcal{R}_\zeta$ has no root in $K_{\mathfrak{p}}$.

Combining the odd-part and pure-power cases proves that

$$\mathcal{R}_\zeta(S) \text{ has no root in } K \qquad (\zeta \neq 1). \tag{75}$$

19 Abelian stability

Proof of Theorem 1.4. Let α be a root of F_ζ and retain the notation $K = \mathbf{Q}(\zeta)$. Theorem 12.2 gives $[K(\alpha) : K] = 6$. Since K^{ab}/K is Galois, the standard degree formula gives

$$[K^{\mathrm{ab}}(\alpha) : K^{\mathrm{ab}}] = [K(\alpha) : K(\alpha) \cap K^{\mathrm{ab}}].$$

Thus the stated intersection equality is equivalent to irreducibility over K^{ab}. Proposition 15.1 supplies an irreducible quartic local factor of F_ζ. Equation (75) proves that the $3+3$ resolvent has no root in K. Theorem 16.1 now proves both assertions. $\square$

20 Generic and torsion-specialized monodromy

The period-4 sextic has full symmetric monodromy generically, and this persists at all but finitely many torsion fibers. Define

$$P_X(T) = T^6 + 2T^5 + 4T^4 + 6T^3 + (X-5)T^2 - 8T - 16. \tag{76}$$

Theorem 20.1 (Generic monodromy)**.** *The geometric and arithmetic Galois groups of P_X over $\overline{\mathbf{Q}}(X)$ and $\mathbf{Q}(X)$ are both S_6. Its splitting field over $\mathbf{Q}(X)$ is regular over $\mathbf{Q}$.*

Proof. The discriminant is

$$\operatorname{disc}_T(P_X) = 2^{10} Q(X), \tag{77}$$

where

$$\begin{aligned} Q(X) = 16X^6 - 619X^5 + 23722X^4 - 366579X^3 \\ + 2799652X^2 - 12598144X + 60081152. \end{aligned}$$

The polynomial Q is squarefree. For a short exact certificate, reduce modulo 5:

$$\overline{Q} = X^6 + X^5 + 2X^4 + X^3 + 2X^2 + X + 2,$$

$$\overline{Q}' = X^5 + 3X^3 + 3X^2 + 4X + 1.$$

The monic nonzero remainders in the Euclidean algorithm are

$$X^4 + 4X + 4, \quad X^3 + 3X^2 + 2, \quad X^2 + 3X, \quad 1.$$

Thus Q has six simple roots; its constant term also shows that none is zero.

The equation $P_X(T) = 0$ is the degree-six rational cover $X = g(T)$ from the period-4 parametrization. Above $X = \infty$ its two poles have orders 4 and 2, so the inertia type is (4)(2) and its ramification contribution is 4. Each of the six finite simple zeros of the discriminant gives a transposition, contributing 6. These contributions total $10 = 2 \cdot 6 - 2$, as required by Riemann–Hurwitz for a degree-six map from $\mathbf{P}^1$.

The cover is connected, and its finite branch cycles are transpositions. Their product is the inverse of the branch cycle at infinity, so they generate the full monodromy group. A transitive permutation group generated by transpositions is the full symmetric group: the transpositions are the edges of a connected graph on the six sheets. Hence the geometric group is S_6. The arithmetic group contains the geometric group and is itself a subgroup of S_6, so it is also S_6. Equality of the two groups is equivalent to regularity of the splitting field. □

Theorem 20.2 (Cofinite torsion specialization)**.** *For all but finitely many roots of unity ζ,*

$$\operatorname{Gal}(P_\zeta/\mathbf{Q}^{\mathrm{ab}}) \cong S_6. \tag{78}$$

For every such nontrivial ζ, if $K = \mathbf{Q}(\zeta)$ and L is the splitting field of P_ζ over K, then

$$\operatorname{Gal}(L/K) = S_6, \qquad L \cap K^{\mathrm{ab}} = K\big(\sqrt{Q(\zeta)}\big), \tag{79}$$

and

$$\operatorname{Gal}(LK^{\mathrm{ab}}/K^{\mathrm{ab}}) = A_6. \tag{80}$$

Proof. Let $\mathcal{L}/\mathbf{Q}(X)$ be the splitting field of P_X. By Theorem 20.1, it is regular and has group $G = S_6$. After removing the branch locus and the points $0, \infty$, its normalization gives a finite étale G-cover over an open subset of $\mathbf{G}_m$. For each conjugacy class of maximal subgroups $M < G$, the fixed field $\mathcal{L}^M$ is the function field of a geometrically integral quotient cover defined over $\mathbf{Q}$; regularity gives

$$(\mathcal{L}^M)\overline{\mathbf{Q}} = (\mathcal{L}\overline{\mathbf{Q}})^M.$$

We claim that every one of these finitely many quotient covers satisfies (PB).

Suppose first that $M \neq A_6$. If (PB) failed, then [Zan10, Proposition 2.1], applied over $\overline{\mathbf{Q}}$, would give a factorization through a nontrivial isogeny of $\mathbf{G}_m$. Thus, for some $n > 1$, the cyclic extension

$$\overline{\mathbf{Q}}(X)(U), \qquad U^n = X,$$

would be contained in $(\mathcal{L}\overline{\mathbf{Q}})^M$. If

$$N = \mathrm{Gal}(\mathcal{L}\overline{\mathbf{Q}}/\overline{\mathbf{Q}}(U)),$$

then $M \subseteq N \lhd G$ and $N \neq G$. Maximality gives $N = M$, so M is normal and G/M is a nontrivial cyclic quotient. The only such maximal kernel in S_6 is A_6, a contradiction. Hence this quotient has (PB).

For $M = A_6$, the quotient is the sign cover

$$W^2 = Q(X).$$

For every $n \geq 1$, the polynomial $Q(U^n)$ has a simple nonzero zero and hence is not a square in $\overline{\mathbf{Q}}(U)$. Thus $W^2 - Q(U^n)$ is irreducible, so this quotient also has (PB).

Apply Zannier's theorem [Zan10, Theorem 2.1] to all these quotient covers simultaneously, and enlarge the finite exceptional set by their branch and bad-specialization points. For a remaining root of unity ζ, let $H_\zeta \leq G$ be the image of $\mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}^{\mathrm{ab}})$ on a geometric fiber of the G-cover. It is the Galois group of P_ζ over $\mathbf{Q}^{\mathrm{ab}}$. The fiber of the M-quotient is irreducible precisely when H_ζ acts transitively on G/M. If H_ζ were proper, it would lie in a conjugate of a maximal subgroup M and would therefore fix a coset in G/M. Since $[G : M] > 1$, this contradicts transitivity. Hence $H_\zeta = G = S_6$, proving (78).

Fix such a nontrivial ζ, put $K = \mathbf{Q}(\zeta)$, and let L be the splitting field over K. Since the Galois group of $L\mathbf{Q}^{\mathrm{ab}}/\mathbf{Q}^{\mathrm{ab}}$ is S_6, the group

$$\mathrm{Gal}(L/L \cap \mathbf{Q}^{\mathrm{ab}})$$

has order 720. It is a subgroup of $\mathrm{Gal}(L/K) \leq S_6$; hence $\mathrm{Gal}(L/K) = S_6$ and $L \cap \mathbf{Q}^{\mathrm{ab}} = K$.

Put $D = L \cap K^{\mathrm{ab}}$. Then D/K is a Galois abelian subextension of the S_6-extension L/K. Its corresponding kernel therefore contains $[S_6, S_6] = A_6$, so

$$D \subseteq L^{A_6}.$$

The discriminant formula (77) identifies L^{A_6} with $K(\sqrt{Q(\zeta)})$. This quadratic extension is abelian over K and hence is contained in D. Thus equality holds, which proves (79); the complementary Galois group over K^{ab} is A_6, proving (80). □

21 The exceptional S_5 resolvent

We now determine the non-solvable core of every nontrivial torsion fiber. Let

$$H = \langle (15364), (16)(24), (3465) \rangle \leq S_6,$$

where the cycles act on the six roots of P_X. This is the exceptional transitive copy of

$$\mathrm{PGL}_2(\mathbf{F}_5) \cong S_5$$

in S_6; it has order 120 and index 6. We use the classical relative invariant recorded in [AFJR15, Table 4]. In variables $x_1, \ldots, x_6$, put

$$\Omega = \sum_{u \in H \cdot (x_1^2 x_2^2 x_3 x_6)} u, \tag{81}$$

where repeated monomials are included only once. The orbit in (81) has 30 elements, and

$$\mathrm{Stab}_{S_6}(\Omega) = H.$$

Both statements are finite: Appendix A describes an exact enumeration of all 720 permutations.

Let $r_1, \ldots, r_6$ be the roots of P_X and take the six coset representatives

$$1, (56), (45), (35), (34), (46).$$

Define

$$\mathcal{C}_X(Z) = \prod_{\sigma \in S_6/H} \big(Z - \sigma\Omega(r_1, \ldots, r_6)\big). \tag{82}$$

Its coefficients are symmetric polynomials in the r_i, so $\mathcal{C}_X \in \mathbf{Z}[X, Z]$. Exact symmetrization gives

$$\begin{aligned}\mathcal{C}_X(Z) = Z^6 - 8A_1(X)Z^5 - 8A_2(X)Z^4 + 8A_3(X)Z^3 \\ + 16A_4(X)Z^2 - 32A_5(X)Z + 16A_6(X),\end{aligned} \tag{83}$$

where

$$\begin{aligned}
A_1 &= X - 25,\\
A_2 &= X^3 - 20X^2 + 705X + 178,\\
A_3 &= 7X^4 - 681X^3 + 16261X^2 - 227443X + 479760,\\
A_4 &= X^6 - 42X^5 + 2702X^4 - 68456X^3 + 1261977X^2\\
&\qquad - 8149990X + 33167184,\\
A_5 &= 3X^7 - 51X^6 + 3210X^5 + 66050X^4 - 1442225X^3\\
&\qquad - 8878943X^2 - 273043148X + 2811612544,\\
A_6 &= 9X^8 + 1042X^7 - 20673X^6 + 266588X^5 - 56301705X^4\\
&\qquad + 359206706X^3 + 7712523489X^2\\
&\qquad - 47208268864X - 339160660992.
\end{aligned}$$

The certificate reconstructs all seven coefficients of (82) directly in the rank-720 ordered splitting algebra of P_X; in particular, the displayed formula is not obtained by numerical approximation.

The discriminant of the new resolvent is

$$\operatorname{disc}_Z(\mathcal{C}_X) = 2^{44} Q(X)^3 J(X)^2, \tag{84}$$

where Q is the polynomial in (77) and

$$\begin{aligned} J(X) = 32X^{11} &- 1878X^{10} + 35932X^9 - 3924653X^8 \\ &+ 205941578X^7 + 2540918305X^6 + 55056832824X^5 \\ &- 3439771613747X^4 + 25424297868354X^3 \\ &+ 117127525549173X^2 - 1681442076883104X \\ &+ 3812315003286784. \end{aligned}$$

For completeness, the coefficient of Z^{6-i} in (83) has X-degree at most

$$1, 3, 4, 6, 7, 8 \qquad (1 \le i \le 6).$$

The discriminant of a monic sextic is weighted homogeneous of weight 30 when that coefficient is assigned weight i. Since the largest ratio of the displayed X-degree to i is $3/2$, the left side of (84) has X-degree at most 45. The exact certificate verifies (84) by 46 integer Sylvester determinants, which proves the polynomial identity.

The constant term of J dominates all the others:

$$|J(0)| - \sum_{i=1}^{11} |[X^i]J| = 1\,984\,823\,523\,717\,204 > 0. \tag{85}$$

Thus $J(z) \neq 0$ whenever $|z| = 1$.

Proposition 21.1 (Squarefree specialization)**.** *For every root of unity $\zeta \neq 1$, the polynomial $\mathcal{C}_\zeta(Z)$ is squarefree over $\mathbf{Q}(\zeta)$. Consequently its irreducible factor degrees are exactly the orbit sizes of the Galois group of F_ζ on S_6/H.*

Proof. Put $K = \mathbf{Q}(\zeta)$. Theorem 12.2 makes F_ζ irreducible over K. In characteristic zero it is therefore separable, so its discriminant identity (77) gives $Q(\zeta) \neq 0$. Equation (85) gives $J(\zeta) \neq 0$, and (84) now proves that $\mathcal{C}_\zeta$ is squarefree.

The six roots in (82) are consequently distinct. The absolute Galois group of K permutes them through its action on the six cosets S_6/H. The irreducible factors of a squarefree polynomial correspond to the orbits on its roots, proving the final assertion. $\square$

22 Local factor degrees of the exceptional resolvent

Let $\zeta \neq 1$ be a root of unity, put $K = \mathbf{Q}(\zeta)$, and fix a prime $\mathfrak{p} \mid 2$. Normalize the valuation on the completion by $v = v_\mathfrak{p}$ and $v(K_\mathfrak{p}^\times) = \mathbf{Z}$, and put $e = v(2)$. Write

$$\operatorname{ord}(\zeta) = 2^a m, \qquad m \text{ odd}.$$

Reduction of the six coefficient polynomials in (83) gives

$$\begin{aligned} &A_1 \equiv X + 1, && A_2 \equiv X(X+1)^2, && A_3 \equiv X(X+1)^3, \\ &A_4 \equiv X^2(X+1)^4, && A_5 \equiv X^2(X+1)^5, && A_6 \equiv X^2(X+1)^6 \pmod 2. \end{aligned} \tag{86}$$

Suppose first that $m > 1$. The reduction of ζ is a nontrivial odd-order root of unity, so both $\bar\zeta$ and $\bar\zeta + 1$ are nonzero. Equation (86) therefore makes every $A_i(\zeta)$ a $\mathfrak{p}$-adic unit. The valuations of the coefficients of $\mathcal{C}_\zeta$, from the constant coefficient to the leading coefficient, are exactly

$$e(4, 5, 4, 3, 3, 3, 0). \tag{87}$$

The lower Newton polygon is the single edge

$$(0, 4e) \longrightarrow (6, 0).$$

Indeed, the five intermediate vertical distances above the endpoint chord are

$$\frac{5e}{3}, \quad \frac{4e}{3}, \quad e, \quad \frac{5e}{3}, \quad \frac{7e}{3}.$$

The slope is $-2e/3$. Since e is a power of two, its reduced denominator is 3.

It remains to treat $m = 1$. Put $\pi = \zeta - 1$. The local cyclotomic facts give

$$v(\pi) = 1, \qquad e = v(2) = 2^{a-1}, \qquad a \geq 1.$$

This includes $\zeta = -1$, for which $\pi = -2$ and $e = 1$. Write

$$A_i(1 + U) = \sum_j a_{ij} U^j.$$

Exact expansion gives the following table; each entry is $v_2(a_{ij})$ and blank entries correspond to absent coefficients.

$i \backslash j$	0	1	2	3	4	5	6	7	8
1	3	0							
2	5	2	0	0					
3	7	3	2	0	0				
4	10	7	5	6	0	2	0		
5	12	8	6	3	5	0	1	0	
6	15	13	11	10	5	3	0	1	0

(88)

In row i, the coefficient a_{ii} is odd, while

$$v_2(a_{ij}) + j > i \qquad (j < i).$$

For $j > i$ the inequality $j > i$ is already strict. Hence

$$v(a_{ij}\pi^j) = e\, v_2(a_{ij}) + j$$

has its unique minimum at $j = i$. It follows that

$$v(A_i(\zeta)) = i \qquad (1 \leq i \leq 6). \tag{89}$$

Thus the coefficient valuations of $\mathcal{C}_\zeta$ are

$$(4e + 6, 5e + 5, 4e + 4, 3e + 3, 3e + 2, 3e + 1, 0). \tag{90}$$

Again the Newton polygon consists of one edge,

$$(0, 4e + 6) \longrightarrow (6, 0).$$

The five vertical distances above its chord are the same positive numbers

$$\frac{5e}{3}, \quad \frac{4e}{3}, \quad e, \quad \frac{5e}{3}, \quad \frac{7e}{3}.$$

Its slope is $-(2e+3)/3$. Since e is a power of two, $3 \nmid 2e+3$, so this slope also has reduced denominator 3.

Proposition 22.1 (Outer-resolvent orbit divisibility)**.** *For every root of unity $\zeta \neq 1$, every irreducible factor of $\mathcal{C}_\zeta$ over $K = \mathbf{Q}(\zeta)$ has degree divisible by* 3*. Equivalently, its factor-degree pattern is either* [6] *or* [3, 3]*. Hence every orbit of* $\mathrm{Gal}(F_\zeta/K)$ *on* S_6/H *has cardinality divisible by* 3*.*

Proof. In both local cases the entire Newton polygon is one edge and the corresponding root valuation has reduced denominator 3. Lemma 14.1 shows that every irreducible factor over $K_\mathfrak{p}$ has degree divisible by 3. A factor over K splits over $K_\mathfrak{p}$ into such local factors, so its degree is also divisible by 3. Since $\mathcal{C}_\zeta$ has degree six, only the two displayed patterns can occur. Proposition 21.1 identifies these factor degrees with the orbit sizes on S_6/H. □

23 Exception-free alternating monodromy

We isolate the finite-group step.

Lemma 23.1 (Alternating criterion for a transitive sextic group)**.** *Let $G \leq S_6$ be transitive. Suppose that*

(i) *every orbit of G on S_6/H has cardinality divisible by* 3*;*

(ii) *G fixes no unordered partition of the six letters into two triples.*

Then $A_6 \leq G$.

Proof. If G has a block of size three, the block and its complement form a fixed unordered $3 + 3$ partition, contrary to (ii). If G has a block of size two, then G is contained in a conjugate of

$$W = S_2 \wr S_3.$$

The group W has orbits of sizes 2 and 4 on S_6/H. Every G-orbit refines a W-orbit, so the orbit of size two contains a G-orbit of size one or two, contrary to (i). Thus G is primitive.

Up to conjugacy, the primitive subgroups of S_6 are

$$\mathrm{PSL}_2(5) \cong A_5, \quad \mathrm{PGL}_2(5) \cong S_5, \quad A_6, \quad S_6;$$

see [BM83]. The first two are contained in a conjugate of the exceptional $S_5 = H$, so each fixes a point of S_6/H. This again contradicts (i). Only A_6 and S_6 remain. The orbit sizes [2, 4], [1, 5], and [6] used here are also checked by the finite permutation certificate in Appendix A. □

Proof of Theorem 1.3. Theorem 12.2 makes the group $G = \mathrm{Gal}(L_\zeta/K)$ transitive. Proposition 22.1 gives hypothesis (i) of Lemma 23.1. A fixed unordered $3 + 3$ partition would give a K-rational root of the resolvent $\mathcal{R}_\zeta$ defined in (65). Equation (75) excludes such a root, so hypothesis (ii) holds. The lemma proves the group-theoretic assertion. The two consequences over K^{sol} stated in Theorem 1.3 are proved in Corollary 24.1 below. □

24 Solvable stability

Fix an algebraic closure $\overline{\mathbf{Q}}$. For a number field $k \subset \overline{\mathbf{Q}}$, write k^{sol} for the compositum in $\overline{\mathbf{Q}}$ of all finite Galois extensions of k with solvable Galois group.

Corollary 24.1 (Exact group and solvable stability)**.** *Under the hypotheses of Theorem 1.3, put*

$$G_\zeta = \mathrm{Gal}(L_\zeta/K), \qquad K_\zeta^{\mathrm{sgn}} = K\big(\sqrt{Q(\zeta)}\big),$$

where the latter field is understood to equal K when $Q(\zeta)$ is a square in K. Then

(i)
$$G_\zeta = \begin{cases} A_6, & Q(\zeta) \in K^{\times 2}, \\ S_6, & Q(\zeta) \notin K^{\times 2}; \end{cases}$$

(ii)
$$L_\zeta \cap K^{\mathrm{sol}} = L_\zeta \cap K^{\mathrm{ab}} = K_\zeta^{\mathrm{sgn}};$$

(iii)
$$\mathrm{Gal}(L_\zeta K^{\mathrm{sol}}/K^{\mathrm{sol}}) \cong \mathrm{Gal}(L_\zeta K^{\mathrm{ab}}/K^{\mathrm{ab}}) \cong A_6;$$

(iv) *F_ζ is irreducible over K^{sol}, and for every root α,*

$$K(\alpha) \cap K^{\mathrm{sol}} = K.$$

Proof. Theorem 1.3 leaves only A_6 and S_6. The discriminant square criterion, together with $\mathrm{disc}(F_\zeta) = 2^{10}Q(\zeta)$, proves (i).

Put $D = L_\zeta \cap K^{\mathrm{sol}}$. Both fields are Galois over K, so D/K is Galois. Its Galois group is a solvable quotient of G_ζ. Since the nonabelian simple group A_6 has no nontrivial solvable quotient, the corresponding kernel contains A_6. Hence

$$D \subseteq L_\zeta^{A_6} = K_\zeta^{\mathrm{sgn}}.$$

The field on the right has degree at most two over K and lies in both L_ζ and K^{sol}, so equality holds. Because

$$K_\zeta^{\mathrm{sgn}} \subseteq K^{\mathrm{ab}} \subseteq K^{\mathrm{sol}},$$

the same inclusions give the asserted intersection with K^{ab}. Restriction then proves (iii).

The natural action of A_6 on six letters is transitive, so F_ζ remains irreducible over K^{sol}. If $E = K(\alpha)$, the standard degree formula over the Galois extension K^{sol}/K gives

$$[K^{\mathrm{sol}}(\alpha) : K^{\mathrm{sol}}] = [E : E \cap K^{\mathrm{sol}}].$$

The left side and $[E : K]$ are both six, so $E \cap K^{\mathrm{sol}} = K$. □

Lemma 24.2 (Comparison of solvable closures)**.** *Let $K/\mathbf{Q}$ be a finite abelian extension. Then*

$$\mathbf{Q}^{\mathrm{sol}} \subseteq K^{\mathrm{sol}}.$$

Moreover, for every $a \in K^\times$,

$$K(\sqrt{a}) \subseteq \mathbf{Q}^{\mathrm{sol}}.$$

Proof. If $M/\mathbf{Q}$ is finite Galois and solvable, then MK/K is finite Galois with Galois group isomorphic to a subgroup of $\mathrm{Gal}(M/\mathbf{Q})$. Thus $M \subseteq K^{\mathrm{sol}}$, and taking the compositum proves the first assertion.

For the second, let N be the compositum of

$$K\big(\sqrt{\sigma(a)}\big), \qquad \sigma \in \mathrm{Gal}(K/\mathbf{Q}).$$

This is the Galois closure of $K(\sqrt{a})/\mathbf{Q}$. The extension N/K is multiquadratic and $K/\mathbf{Q}$ is abelian, so $\mathrm{Gal}(N/\mathbf{Q})$ is solvable. Hence $N \subseteq \mathbf{Q}^{\mathrm{sol}}$. □

Corollary 24.3 (Monodromy over the maximal solvable extension)**.** *With the notation above,*

$$L_\zeta \cap \mathbf{Q}^{\mathrm{sol}} = K_\zeta^{\mathrm{sgn}}, \qquad \mathrm{Gal}(L_\zeta \mathbf{Q}^{\mathrm{sol}}/\mathbf{Q}^{\mathrm{sol}}) \cong A_6.$$

In particular, F_ζ is irreducible over $\mathbf{Q}^{\mathrm{sol}}$.

Proof. The cyclotomic field K is contained in $\mathbf{Q}^{\mathrm{sol}}$. The preceding lemma and Corollary 24.1 give

$$K_\zeta^{\mathrm{sgn}} \subseteq L_\zeta \cap \mathbf{Q}^{\mathrm{sol}} \subseteq L_\zeta \cap K^{\mathrm{sol}} = K_\zeta^{\mathrm{sgn}}.$$

Restriction gives the group A_6, whose natural action is transitive. □

25 Factorization over fields with solvable Galois closure

For $k \geq 2$, recall the fixed-degree product

$$\Gamma_k(T) = \prod_{\substack{\zeta^k=1 \\ \zeta \text{ primitive}}} F_\zeta(T). \tag{91}$$

Theorem 25.1 (Exact solvable-base factorization)**.** *Let $k \geq 2$, put $K_k = \mathbf{Q}(\zeta_k)$, and let $B/\mathbf{Q}$ be any finite extension whose Galois closure over $\mathbf{Q}$ is solvable. Set*

$$D = B \cap K_k.$$

Then Γ_k has exactly $[D : \mathbf{Q}]$ irreducible factors over B, each of degree $6[K_k : D]$. Equivalently, the factors are indexed by the $\mathrm{Gal}(\overline{\mathbf{Q}}/B)$-orbits on the primitive k-th roots, and the factor attached to an orbit $\mathcal{O}$ is

$$\Gamma_{\mathcal{O}}(T) = \prod_{\zeta \in \mathcal{O}} F_\zeta(T).$$

In particular,

$$\Gamma_k \text{ is irreducible over } B \iff B \cap K_k = \mathbf{Q}.$$

Proof. Because $K_k/\mathbf{Q}$ is Galois, restriction identifies the image of $\mathrm{Gal}(\overline{\mathbf{Q}}/B)$ in $\mathrm{Gal}(K_k/\mathbf{Q})$ with $\mathrm{Gal}(K_k/D)$. Equivalently,

$$\mathrm{Gal}(BK_k/B) \cong \mathrm{Gal}(K_k/D).$$

The action of $\mathrm{Gal}(K_k/\mathbf{Q})$ on the primitive k-th roots is simply transitive. Hence there are $[D : \mathbf{Q}]$ relevant orbits, each of cardinality $[K_k : D]$.

Let $\widetilde{B}/\mathbf{Q}$ be the Galois closure of $B/\mathbf{Q}$. The extension $\widetilde{B}K_k/K_k$ is finite Galois with solvable Galois group, so

$$BK_k \subseteq \widetilde{B}K_k \subseteq K_k^{\mathrm{sol}}.$$

Corollary 24.1 therefore shows that every F_ζ remains irreducible over BK_k.

Fix an orbit $\mathcal{O}$, choose $\zeta \in \mathcal{O}$, and let α be a root of F_ζ. Since $F_\zeta(0) = -16$, one has $\alpha \neq 0$, and $\zeta = g(\alpha)$ gives

$$B(\alpha) = BK_k(\alpha).$$

Consequently

$$[B(\alpha) : B] = [BK_k(\alpha) : BK_k]\,[BK_k : B] = 6[K_k : D].$$

The polynomial $\Gamma_{\mathcal{O}}$ belongs to $B[T]$, is monic, has degree $6|\mathcal{O}| = 6[K_k : D]$, and vanishes at α. It is therefore the minimal polynomial of α over B. Different orbit products are coprime: a common root α would make $g(\alpha)$ belong to two disjoint orbits. This proves that the displayed products give the complete factorization. □

Corollary 25.2 (Complete factorization over the maximal solvable extension)**.** *For every $k \geq 2$, equation (91) is the complete irreducible factorization of Γ_k over $\mathbf{Q}^{\mathrm{sol}}$. It consists of $\varphi(k)$ distinct sextics, and the splitting field of each sextic over $\mathbf{Q}^{\mathrm{sol}}$ has Galois group A_6. The same product is also the complete factorization over $\mathbf{Q}^{\mathrm{ab}}$.*

Proof. Every primitive k-th root belongs to $\mathbf{Q}^{\mathrm{sol}}$, and Corollary 24.3 makes each F_ζ irreducible over that field with splitting group A_6. Distinct primitive roots give distinct sextics because their T^2 coefficients are $\zeta - 5$. Equation (91) proves completeness. Irreducibility over the larger field $\mathbf{Q}^{\mathrm{sol}}$ also implies irreducibility over $\mathbf{Q}^{\mathrm{ab}} \subseteq \mathbf{Q}^{\mathrm{sol}}$. □

26 Solvable-base factorization of the delta factors

Lemma 13.2 gives a $\mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$-equivariant bijection

$$\beta : Z(\Gamma_k) \longrightarrow Z(\widetilde{\Delta}_{4k,4}), \qquad \beta(\alpha) = h(\alpha).$$

Theorem 26.1 (Exact solvable-base delta factorization)**.** *Let $k \geq 2$ and let $B/\mathbf{Q}$ be a finite extension whose Galois closure is solvable. With $K_k = \mathbf{Q}(\zeta_k)$ and $D = B \cap K_k$, the polynomial $\widetilde{\Delta}_{4k,4}$ has exactly $[D : \mathbf{Q}]$ irreducible factors over B, each of degree $6[K_k : D]$. Over $\mathbf{Q}^{\mathrm{sol}}$ its complete factorization consists of $\varphi(k)$ irreducible sextic factors, one for each primitive k-th root. The splitting group of each such sextic over $\mathbf{Q}^{\mathrm{sol}}$ is A_6.*

Proof. Restricting the equivariance of β to $\mathrm{Gal}(\overline{\mathbf{Q}}/B)$ identifies its orbits, including their cardinalities. Theorem 25.1 gives the assertion over B, and Corollary 25.2 gives the assertion over $\mathbf{Q}^{\mathrm{sol}}$. On each sextic orbit the two permutation representations are isomorphic, so their kernels and hence their splitting fields agree; the group is therefore A_6. □

The same assertions hold for the unscaled factor $\Delta_{4k,4}(c)$, because $C = 4c$ is an invertible linear change over $\mathbf{Q}$.

Remark 26.2. *The preceding theorem is an orbit description; it does not identify the sextic in the parameter variable with $F_\zeta(T)$.*

Corollary 26.3 (Coprime cyclotomic base change)**.** *Let $k \geq 2$ and $\ell \geq 1$, and let ζ_ℓ be a primitive ℓ-th root of unity. If $(k, \ell) = 1$, then both Γ_k and $\widetilde{\Delta}_{4k,4}$ remain irreducible over $\mathbf{Q}(\zeta_\ell)$.*

Proof. For coprime k and ℓ, $K_k \cap \mathbf{Q}(\zeta_\ell) = \mathbf{Q}$. Apply Theorems 25.1 and 26.1. □

Remark 26.4. *Corollary 26.3 is the period-4 analogue of the period-3 result in [KMST25, Theorem 1.2]. The theorems above are stronger: they allow every finite base field with solvable Galois closure and describe every resulting factor.*

27 The solvable part of a parameter field

Corollary 27.1 (Maximal solvable subfield of a parameter field)**.** *Let $C = 4c$ be an exact-period-4 parabolic parameter with primitive k-th root multiplier ζ, where $k \geq 2$. Then*

$$\mathbf{Q}(C) \cap \mathbf{Q}^{\mathrm{sol}} = \mathbf{Q}(\zeta).$$

Equivalently, $\mathbf{Q}(\zeta)$ is the largest subfield of $\mathbf{Q}(C)$ whose Galois closure over $\mathbf{Q}$ is solvable. In particular, $C \notin \mathbf{Q}^{\mathrm{sol}}$. The same assertions hold with c in place of C.

Proof. Let α be the corresponding root of F_ζ. The injective Galois-equivariant root correspondence gives

$$\mathbf{Q}(C) = \mathbf{Q}(\alpha) = \mathbf{Q}(\zeta, \alpha).$$

Put $K = \mathbf{Q}(\zeta)$ and $E = K(\alpha)$. Corollary 24.1 gives

$$E \cap K^{\mathrm{sol}} = K.$$

By Lemma 24.2, $\mathbf{Q}^{\mathrm{sol}} \subseteq K^{\mathrm{sol}}$, while $K \subseteq E \cap \mathbf{Q}^{\mathrm{sol}}$. Therefore

$$K \subseteq E \cap \mathbf{Q}^{\mathrm{sol}} \subseteq E \cap K^{\mathrm{sol}} = K.$$

If $M \subseteq E$ has solvable Galois closure over $\mathbf{Q}$, then $M \subseteq \mathbf{Q}^{\mathrm{sol}}$ and hence $M \subseteq K$. Conversely, $K/\mathbf{Q}$ is abelian. Thus K is the largest such subfield. If $C \in \mathbf{Q}^{\mathrm{sol}}$, then $E = \mathbf{Q}(C) \subseteq \mathbf{Q}^{\mathrm{sol}}$, contradicting $[E : K] = 6$. Finally $C = 4c$, so the two coordinates generate the same field. □

Corollary 27.2 (Maximal Galois subfield)**.** *Under the hypotheses of Corollary 27.1, the cyclotomic field $\mathbf{Q}(\zeta)$ is the largest subfield of $\mathbf{Q}(C)$ that is Galois over $\mathbf{Q}$.*

Proof. Write $K = \mathbf{Q}(\zeta)$ and $E = \mathbf{Q}(C) = K(\alpha)$. The proof of Theorem 16.1 gives the stronger fact that E contains no nontrivial intermediate field Galois over K. If $D \subseteq E$ is Galois over $\mathbf{Q}$, then DK is contained in E and is Galois over K. Hence $DK = K$, so $D \subseteq K$. Conversely, $K/\mathbf{Q}$ is Galois. □

At multiplier 1, the exact-period-4 parameters instead satisfy

$$p(C) = C^3 + 9C^2 + 27C + 135 = 0.$$

After $U = C + 3$, this is $U^3 + 108$. The cubic is irreducible over $\mathbf{Q}$, and its discriminant is

$$-27 \cdot 108^2 = -2^4 3^9,$$

which is not a square. Its Galois closure has group S_3. Consequently these parameters do not lie in $\mathbf{Q}^{\mathrm{ab}}$, but they do lie in $\mathbf{Q}^{\mathrm{sol}}$. This explains why the solvable parameter-field statement above necessarily assumes $k \geq 2$.

28 Torsion Hilbert irreducibility context

For this interpretation one must use the finite cover of degree six

$$g : U \longrightarrow \mathbf{G}_m, \qquad U = \mathbf{G}_m \setminus g^{-1}(0).$$

Its degree is the degree of the rational function g, namely six. After writing the target coordinate as $X = g(T)$, the element T satisfies the monic equation $P_X(T) = 0$ with nonzero constant term; both T and T^{-1} are integral over $\overline{\mathbf{Q}}[X, X^{-1}]$, which also proves finiteness. The pullback condition means that the fiber product with $x \mapsto x^n$ is geometrically irreducible for every $n \geq 1$. Its function-field equation is

$$Y^n = g(T).$$

Over $\overline{\mathbf{Q}}(T)$, the function g is not a p-th power for any prime p. For $p \geq 3$, this follows from its pole of order two at $T = 0$. For $p = 2$, suppose $g = w^2$. Its only poles force

$$w = aT^2 + bT + c + dT^{-1}.$$

Comparison of the coefficients of T^4, T^3, T^{-1}, T^{-2} gives

$$a^2 = -1, \qquad b = a, \qquad d^2 = 16, \qquad cd = 4.$$

Thus $c^2 = 1$, whereas comparison at T^2 gives

$$-1 + 2ac = -4, \qquad ac = -\frac{3}{2},$$

so $(ac)^2 = 9/4$, contradicting $a^2c^2 = -1$. Hence g is not a square.

The field $\overline{\mathbf{Q}}(T)$ contains all roots of unity. The Kummer binomial criterion [Lan02, Chapter VI, §9, Theorem 9.1] says in this setting that $Y^n - q$ is irreducible if and only if q is not a p-th power for every prime $p \mid n$. In the general criterion one must also exclude $q \in -4\overline{\mathbf{Q}}(T)^4$ when $4 \mid n$; here every element $-4b^4 = (2ib^2)^2$ is already a square. The preceding arguments therefore prove that $Y^n - g(T)$ is irreducible for every $n \geq 1$.

Zannier's cyclotomic Hilbert irreducibility theorem [Zan10, Theorem 2.1] then gives finitely many exceptional torsion fibers over the cyclotomic closure. By Kronecker–Weber this closure is $\mathbf{Q}^{\mathrm{ab}}$. Corollary 24.1 determines the exceptions exactly: for $\zeta \neq 1$, the polynomial F_ζ is irreducible even over $\mathbf{Q}(\zeta)^{\mathrm{sol}}$, while

$$F_1(T) = (T + 2)(T^2 + 4)(T^3 - 2).$$

Thus $\zeta = 1$ is the unique exceptional torsion fiber for this cover.

A Exact symbolic certificates

The verification directory contains dependency-free exact certificates using integer polynomial arithmetic. The principal program constructs the companion matrix of P_X, constructs the matrix of its third exterior power from all 3×3 minors, and verifies every coefficient of the exterior-power resolvent identity and of the displayed resolvent formula. It also checks the three polynomial congruences and the parametrization identity (60) coefficient by coefficient. Finally, it certifies (77) from twelve exact Sylvester determinants under the elementary degree bound and verifies the displayed Euclidean algorithm modulo 5.

The outer-resolvent certificate enumerates all 720 permutations of six letters, verifies the order-120 stabilizer of the exceptional invariant and the required coset-orbit table, and reconstructs every coefficient of $\mathcal{C}_X$ directly in the rank-720 ordered splitting algebra of P_X. It verifies (84) from 46 exact Sylvester determinants under the proved degree-45 bound, and checks the unit-circle estimate, the six congruences (86), and every entry of the Taylor table (88). The programs exit with a nonzero status on any failed equality or nonexact integer division. No floating-point or inexact numerical computation occurs.

From the directory containing this manuscript, run

```
python3 verification/verify_all.py
```

Declarations

Declaration on artificial-intelligence assistance. OpenAI GPT-5.6 Sol Pro was used to assist in developing the core period-4 irreducibility argument and its initial abelian-stability extension. OpenAI GPT-5.6 Sol Ultra, accessed through Codex, was used to assist with further mathematical development, line-by-line proof checking, preparation of the exact-arithmetic verification code, and drafting and editing the manuscript. Anthropic Claude Opus 5 was used in a limited number of desktop-chat exchanges for additional critical review. All AI-assisted arguments, citations, and computations were subjected to repeated line-by-line verification. The author takes full responsibility for the accuracy, integrity, and originality of the article.

Data and code availability. The exact-arithmetic verification code supporting the computer-assisted certificates in Appendix A is available at the following versioned address:

https://github.com/Isaac-Wei-2026/
morton-vivaldi-period-4/tree/v3.4/final-paper/verification

The programs use dependency-free Python 3 and exact integer arithmetic. No other research data were generated or analysed.

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